

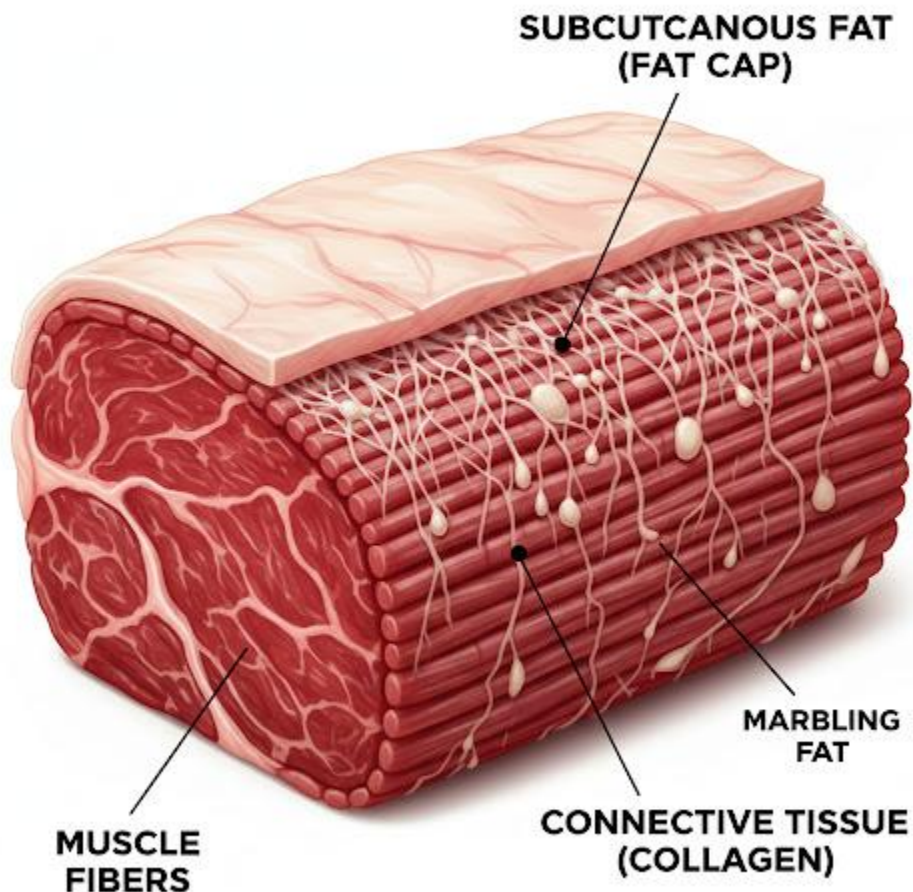
SS2 FOOD AND NUTRITION LESSON NOTES

FIRST TERM

WEEK 1: MEAT COOKERY

TOPIC: MEAT COOKERY

INTRODUCTION: Meat is a primary source of high-quality protein and essential nutrients in the human diet. Understanding its structure, types, and cookery is fundamental to food and nutrition. This week, we will conduct an in-depth exploration of meat, from its biological structure to its culinary applications.



1: MEANING AND TYPES OF MEAT

- **MEANING:** Meat is defined as the flesh (muscle tissue) of animals, primarily mammals and birds, consumed as food. It is a staple in many cultures and is valued for its protein content and flavour.
- **TYPES OF MEAT:** Meat is broadly classified based on the animal source and the colour of the flesh, which is determined by the concentration of myoglobin (an oxygen-storing protein) in the muscle.

1. RED MEAT:

- **SOURCE:** Typically from mammals.
- **EXAMPLES:** Beef (from adult cattle), Veal (from young cattle), Lamb (from young sheep), Mutton (from adult sheep), Pork (from pigs), Goat meat.
- **CHARACTERISTICS:** Higher in myoglobin, which gives it a characteristic red colour when raw. Often has a stronger flavour and is a rich source of iron and zinc.

2. WHITE MEAT:

- **SOURCE:** Typically from poultry (birds) and some cuts of pork.
- **EXAMPLES:** Chicken, Turkey, Duck (though the leg meat is darker), Rabbit.
- **CHARACTERISTICS:** Lower in myoglobin, resulting in a paler colour. Generally has a milder flavour and is often leaner than red meat.

3. OFFAL (VARIETY MEATS):

- **MEANING:** These are the edible internal organs and other parts of the animal, excluding muscle.
- **EXAMPLES:** Liver, Kidney, Heart, Tongue, Tripe (stomach lining), Brain, Sweetbreads (thymus gland).
- **CHARACTERISTICS:** Often highly nutritious, providing concentrated sources of vitamins (like A and B12) and minerals, but also high in cholesterol. They have unique textures and flavours.

2: STRUCTURE OF MEAT

Understanding the structure of meat is crucial because it directly determines the texture, flavour, and appropriate cooking method. Meat is composed of three main types of tissue:

- **1. MUSCLE TISSUE (THE FLESH):**

- This is the main component we eat.
- It is made of bundles of long, thin muscle fibres.
- **TEXTURE:** The *grain* of the meat refers to the direction of these fibres. The size of the fibres determines the texture.
 - **Fine-grained:** Smaller fibres, found in less-used muscles (e.g., the loin, tenderloin). These cuts are naturally tender.
 - **Coarse-grained:** Larger fibres, found in heavily-used muscles (e.g., the leg, shoulder, neck). These cuts are tougher.

- **2. CONNECTIVE TISSUE:**

- This is the "glue" that holds the muscle fibres together in their bundles and attaches the muscle to the bone.
- It is the *primary factor* in determining meat toughness.
- **TYPES:**
 - **COLLAGEN (WHITE):** Very tough, but when cooked slowly with moisture (e.g., stewing, braising), it breaks down, melts, and turns into gelatin. This makes the meat tender, moist, and gives the sauce a rich body. (Found in tough cuts like shoulder).
 - **ELASTIN (YELLOW):** Extremely tough and *does not* break down with cooking. It must be physically removed (trimmed) or tenderized by pounding or cutting. (Found in "gristle" and ligaments).

- **3. FATTY TISSUE (ADIPOSE TISSUE):**

- Fat is a critical component for flavour, moisture, and tenderness.
- **TYPES:**
 - **SUBCUTANEOUS FAT:** The thick layer of fat on the *outside* of the cut, just under the skin. It can be trimmed off.
 - **INTRAMUSCULAR FAT (MARBLING):** Specks or streaks of fat *within* the muscle bundles. This is highly desirable in cuts for roasting or

grilling, as it melts during cooking, basting the meat from the inside and adding flavour and juiciness.

3: NUTRITIVE VALUE OF MEAT

- **PROTEIN (HIGH BIOLOGICAL VALUE):** Meat is a "complete" protein, meaning it contains all nine essential amino acids required for human growth, repair, and immune function.
- **FAT:** Provides a concentrated source of energy. The *type* of fat varies. Red meat tends to be higher in saturated fat, while some cuts are leaner.
- **MINERALS:**
 - **IRON:** Meat (especially red meat) is the best source of *haem iron*, which is much more easily absorbed by the body than non-haem iron (from plants). Essential for preventing anaemia.
 - **ZINC:** Crucial for immune function and wound healing.
 - **PHOSPHORUS:** Works with calcium for bone health.
- **VITAMINS:**
 - **B-COMPLEX VITAMINS:** Meat is a major source, especially **Vitamin B12** (which is almost exclusively found in animal products), Niacin (B3), and Riboflavin (B2). These are vital for energy metabolism and nervous system function.
 - **Vitamin A:** Found in very high concentrations in offal, especially liver.

4: FACTORS AFFECTING CHOICE OF MEAT

1. **THE CUT:** This is the most important factor. The "cut" refers to the part of the animal the meat comes from.
 - **TENDER CUTS:** From less-used muscles (e.g., loin, rib). Best for quick, *dry-heat* cooking (grilling, roasting). More expensive.
 - **TOUGH CUTS:** From heavily-used muscles (e.g., shoulder, leg, shank). More connective tissue. Best for slow, *moist-heat* cooking (stewing, braising). Less expensive, but more flavourful.

2. **AGE OF ANIMAL:** Younger animals (veal, lamb) are generally more tender and have a milder flavour than older animals (beef, mutton).
3. **MARBLING:** More marbling (intramuscular fat) indicates a cut that will be more tender, juicy, and flavourful.
4. **FRESHNESS:** Look for bright, appealing colour (beef should be cherry-red, pork pinkish-grey), firm texture, and no "off" or sour smell.
5. **BUDGET (INCOME):** Tender cuts are expensive. Tough cuts, offal, and mince are more economical.
6. **COOKING METHOD:** The chosen cooking method dictates the cut (e.g., you don't grill a tough "stewing" cut).
7. **FAMILY SIZE/PREFERENCES:** The amount needed and the family's tastes.

5: METHODS OF COOKING MEAT AND MEAT DISHES

- The goal is to make the meat tender, flavourful, and safe to eat (killing harmful bacteria).
- **1. DRY-HEAT METHODS (FOR TENDER CUTS):**
 - **ROASTING:** Cooking by dry heat in an oven (e.g., roast beef, whole chicken).
 - **GRILLING (OR BROILING):** Cooking by radiant heat from above or below (e.g., steak, suya).
 - **FRYING:**
 - **Shallow Frying:** Cooking in a small amount of fat (e.g., sausages, burger patties).
 - **Deep Frying:** Submerging the meat in hot fat (e.g., fried chicken).
 - **STIR-FRYING:** Frying small pieces of meat very quickly in a wok with minimal oil (e.g., beef stir-fry).
- **2. MOIST-HEAT METHODS (FOR TOUGH CUTS):**
 - **STEWING:** Cooking small pieces of meat slowly in liquid (stock, water, wine) in a pot with a lid. The meat is fully submerged. (e.g., Nigerian beef stew, goat meat stew).
 - **BRAISING:** Cooking large pieces of meat slowly in a small amount of liquid in a covered pot (pot-roasting). The meat is *not* fully submerged. This breaks down collagen.

- **BOILING/SIMMERING:** Cooking in water. Often used to pre-cook meat for stews or to make stock.
- **MEAT DISHES:**
 - **STEWs:** Beef stew, Goat meat stew.
 - **SOUPS:** Oxtail soup, Pepper soup.
 - **GRILLED:** Suya, Asun (smoked goat meat).
 - **ROASTED:** Roast beef, Roast chicken.
 - **MINCE DISHES:** Meatballs, Bolognese sauce, Meat pies.

EVALUATION

1. What is the primary difference between red meat and white meat? Give two examples of each.
2. Explain the three main components of meat's structure.
3. What is the difference between collagen and elastin? How does each one affect your choice of cooking method?
4. Why is red meat (like beef) considered crucial for preventing anaemia?
5. You have a very tough, but flavourful, cut of beef (like the shoulder). Which cooking method (grilling or stewing) would you choose, and *why*?

ASSIGNMENT

1. Draw a large diagram illustrating the structure of meat. Label the muscle fibres, muscle bundles, connective tissue (collagen), and both subcutaneous and intramuscular fat (marbling).
2. Visit a local market or butcher. Identify and list five different cuts of meat (e.g., "beef shoulder," "goat leg," "oxtail"). For each, classify it as "tough" or "tender" and state the *best* cooking method for it.
3. What is "offal"? Write a 400-word report on the nutritive value (both pros and cons) of consuming liver.
4. Create a two-column table. In Column A, list all the dry-heat cooking methods. In Column B, list all the moist-heat cooking methods.

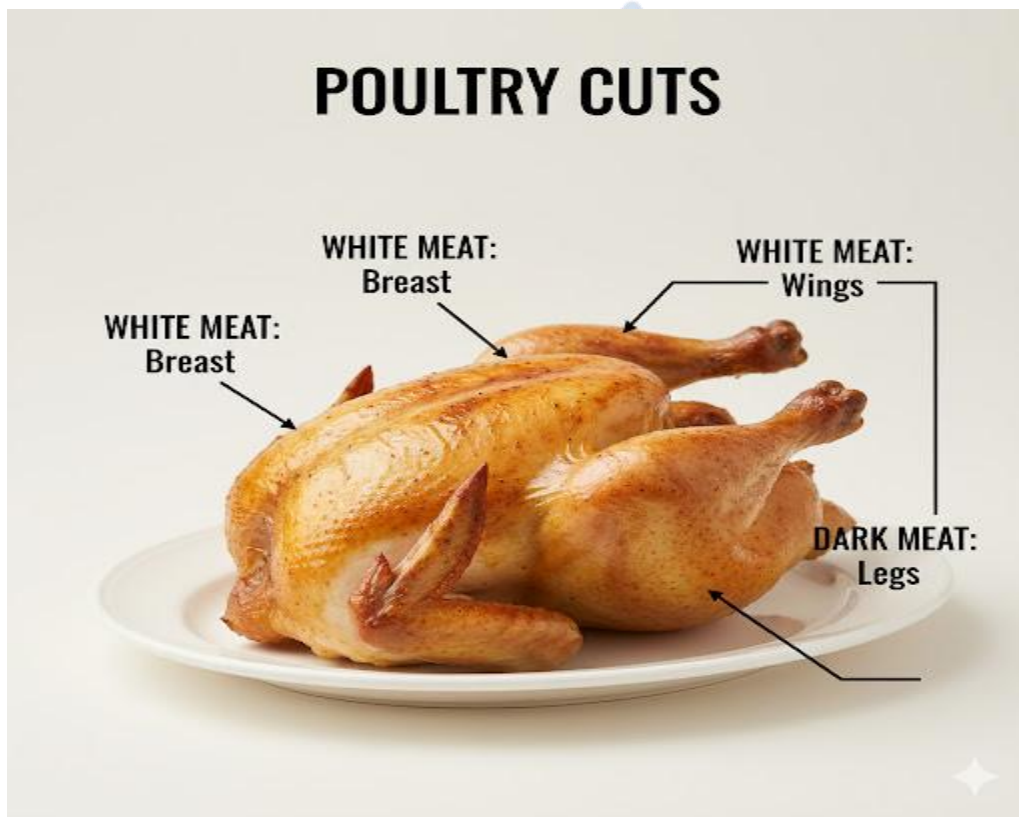
5. Research the process of "tenderizing" meat *before* cooking. Describe three different methods (e.g., mechanical, chemical, enzymatic).



WEEK 2: POULTRY

TOPIC: POULTRY COOKERY

INTRODUCTION: Poultry refers to domesticated birds farmed for their meat, eggs, and feathers. Like meat, poultry is a versatile and globally popular source of high-quality protein. This week, we will explore the different types of poultry, their preparation, nutritional benefits, and cooking methods.



1: MEANING AND TYPES OF POULTRY

- **MEANING:** Poultry includes all domesticated birds raised for human consumption.
- **TYPES OF POULTRY (COMMON):**
 1. **CHICKEN:** The most common type.
 - **Broiler/Fryer:** Young chicken (6-12 weeks), very tender, suitable for all cooking methods.
 - **Roaster:** Older chicken (3-5 months), larger, good for roasting.

- **Hen/Stewer (Old Layer):** Older chicken (>10 months), much tougher, high in fat, and very flavourful. Must be cooked with slow, moist-heat methods (e.g., stews, stock). This is the common "local" chicken.
- 2. **TURKEY:** Larger than chicken, with leaner breast meat and darker, richer leg meat.
- 3. **DUCK:** Has a thick layer of fat under the skin and dark, rich, flavourful meat. Best cooked with dry heat (like roasting) to render the fat.
- 4. **GOOSE:** A large bird, very high in fat, typically reserved for festive occasions.
- 5. **GUINEA FOWL:** Common in Nigeria, it has a "gamier" (stronger) flavour than chicken and is leaner.
- 6. **PIGEON (SQUAB):** Young pigeon, considered a delicacy, has dark, tender meat.

2: STRUCTURE AND NUTRITIVE VALUE

- **STRUCTURE:**

- The structure is similar to meat: muscle, connective tissue, and fat.
- **MUSCLE:** Poultry has two distinct types of meat:
 - **WHITE MEAT (Breast & Wings):** Less-used muscles, less myoglobin, very lean, tender. Prone to drying out if overcooked.
 - **DARK MEAT (Legs & Thighs):** More-used muscles, more myoglobin and fat, more flavourful, and juicier. Takes longer to cook. (Note: Flight birds like duck and goose have dark meat almost everywhere).
- **CONNECTIVE TISSUE:** Less connective tissue than red meat, which is why poultry is generally more tender.
- **FAT:** Stored in the skin, in the abdominal cavity, and just under the skin (not in marbling). This is why the skin can be removed to reduce fat content.

- **NUTRITIVE VALUE:**

- **PROTEIN:** Excellent source of high-biological-value protein.
- **FAT:** Generally low in fat, *especially* if the skin is removed. The fat is also less saturated than beef fat.
- **VITAMINS:** Good source of B-complex vitamins, especially Niacin (B3).

- **MINERALS:** Contains iron and zinc, though in lower amounts than red meat (dark meat is a better source than white meat).

3: PREPARATION OF POULTRY

1. **PURCHASING:** Can be bought live, freshly-dressed (killed and plucked), or frozen.
2. **DRESSING (IF LIVE):** Involves killing, scalding (dipping in hot water to loosen feathers), plucking (removing feathers), and eviscerating (removing internal organs).
3. **WASHING:** Wash the bird thoroughly, inside and out, under cold running water.
4. **DISJOINTING (CUTTING):** Cutting the bird into its principal parts (e.g., 2 legs, 2 thighs, 2 wings, 2 breast halves).
5. **TRUSSING:** Tying a whole bird (especially chicken or turkey) with kitchen string into a compact shape.
 - **PURPOSE:** Ensures even cooking. It stops the wings and legs from sticking out and burning before the breast is cooked.
6. **MARINATING:** Soaking the poultry in a seasoned liquid (a marinade) to add flavour and moisture.
7. **FOOD SAFETY (CRITICAL):**
 - Raw poultry is a high-risk food. It often carries **Salmonella** bacteria.
 - **NEVER** wash raw chicken in a way that splashes water around your sink. This cross-contaminates your entire kitchen. Pat it dry with paper towels instead.
 - Use a separate chopping board for raw poultry.
 - Wash your hands, the board, and any knives thoroughly with hot, soapy water immediately after contact.

4: FACTORS AFFECTING CHOICE OF POULTRY

1. **FRESHNESS:** If buying fresh, look for a plump breast, moist (not dry) skin, and no "off" smell.
2. **AGE/TYPE:** The type of bird and its age determine the cooking method. A young broiler is for frying; an old layer (hen) is for stewing.

3. **INTENDED USE:** Are you making a stew (buy an old hen), a roast (buy a roaster), or fried chicken (buy a broiler)?
4. **BUDGET:** A whole chicken is often cheaper per kilo than pre-cut pieces. Turkey is more expensive.
5. **NUTRITIONAL GOALS:** A person on a low-fat diet will choose a skinless chicken breast.

5: METHODS OF COOKING POULTRY AND POULTRY DISHES

- Poultry is extremely versatile and can be cooked by almost any method.
- **DRY-HEAT METHODS (BEST FOR YOUNG BIRDS/PIECES):**
 - **ROASTING:** The classic method for a whole chicken, turkey, or duck.
 - **GRILLING/BARBECUING:** Excellent for chicken pieces (wings, legs).
 - **FRYING (SHALLOW, DEEP, STIR-):** Very common (e.g., deep-fried chicken, chicken stir-fry).
- **MOIST-HEAT METHODS (BEST FOR OLD, TOUGH BIRDS):**
 - **STEWING/BRAISING:** Ideal for "old layers" or "local" chicken, as the slow, moist heat breaks down the tougher muscle and creates a rich stew.
 - **BOILING/SIMMERING:** Used to cook chicken for soups, salads, or to make chicken stock.
- **POULTRY DISHES:**
 - Chicken Stew
 - Jollof Rice with Chicken
 - Chicken Pepper Soup
 - Roast Turkey (Festive)
 - Fried Chicken
 - Chicken and Mushroom Pie
 - Chicken Suya

EVALUATION

1. What is the main difference between "white meat" and "dark meat" in poultry?

2. Why is an "old layer" (hen) unsuitable for roasting, and which cooking method is best for it?
3. What is "trussing," and what is its main purpose?
4. What is the *most critical* food safety risk associated with raw poultry, and how can it be managed in the kitchen?
5. List three types of poultry (other than chicken) and a suitable cooking method for one of them.

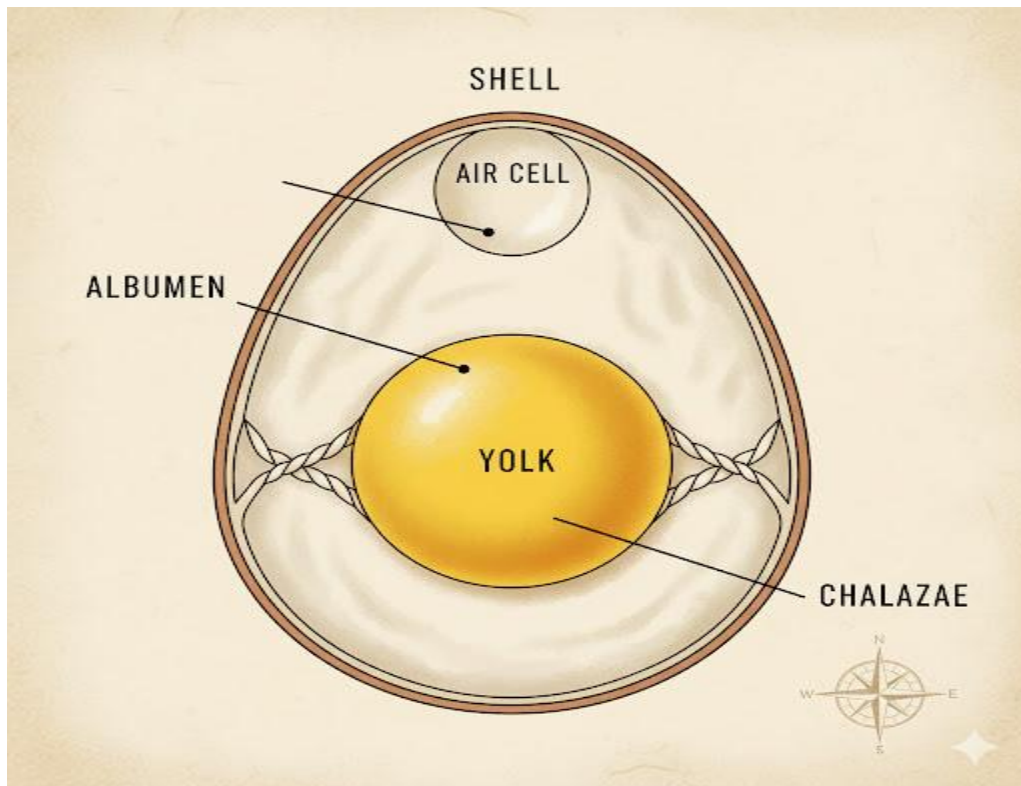
ASSIGNMENT

1. Write a 400-word essay comparing the nutritive value of a skinless chicken breast (white meat) versus a chicken thigh (dark meat).
2. Research the step-by-step process of "dressing" a live chicken. List all the steps from killing to eviscerating.
3. What is a "marinade"? Create a recipe for a chicken marinade using common Nigerian ingredients (e.g., ginger, garlic, pepper, stock cubes).
4. Go to a market or cold room. Find the price for a whole frozen chicken, a whole "local" chicken (old layer), and pre-cut chicken wings. Create a table and compare the cost-effectiveness.
5. Draw a simple diagram of a whole chicken and label the 8 main pieces you get from "disjointing" it.

WEEK 3: EGGS

TOPIC: EGG COOKERY

INTRODUCTION: Eggs are one of nature's most complete and versatile foods. Produced by female birds (most commonly chickens), they are a powerhouse of nutrition and serve a vast number of functions in cookery, from binding and thickening to leavening and glazing.



1: STRUCTURE OF AN EGG

To understand how to cook an egg, we must first understand its structure.

1. **SHELL:** The hard, protective outer layer. It is porous (has tiny holes), which allows moisture and air to pass through (this is why eggs can absorb smells).
2. **SHELL MEMBRANES:** Two thin membranes (inner and outer) just inside the shell. They act as a barrier against bacteria.

3. **AIR CELL (AIR POCKET):** A pocket of air, usually at the "blunt" end of the egg, found between the two membranes. As an egg ages, it loses moisture, and the air cell gets *larger*.
4. **ALBUMEN (EGG WHITE):**
 - This is the clear, thick liquid that surrounds the yolk.
 - It is about 90% water and 10% protein (mainly albumin).
 - It comes in two layers: a thick, viscous white and a thin, watery white.
5. **YOLK (EGG YOLK):**
 - The yellow/orange centre.
 - This is the main nutritive part of the egg, containing all the fat, cholesterol, and most of the vitamins and minerals.
 - It also contains protein and *lecithin*, an emulsifier.
6. **CHALAZAE (PRONOUNCED 'KA-LA-ZE'):** Two "ropy," spiral strands of albumen that anchor the yolk in the centre of the thick white. They are a sign of freshness and are edible.

2: NUTRITIVE VALUE OF EGGS

- **PROTEIN:** An egg contains about 6-7 grams of "complete" protein (High Biological Value), serving as the gold standard for protein quality.
- **FAT:** All fat is in the yolk. It is a mix of saturated and unsaturated fats.
- **CHOLESTEROL:** The yolk is high in dietary cholesterol.
- **VITAMINS:**
 - Rich in **Vitamin D** (one of the few food sources).
 - Rich in **Vitamin A, E, K** (fat-soluble, in the yolk).
 - Rich in **B-Complex Vitamins**, especially **B12** and **Folate**.
- **MINERALS:**
 - Rich in **Iron** (in the yolk).
 - Rich in **Phosphorus** and **Zinc**.
- **CARBOHYDRATES:** Eggs contain virtually no carbohydrates.

3: CULINARY USES OF EGGS

Eggs are indispensable in the kitchen for their unique chemical properties:

1. **BINDING:** The proteins in the egg coagulate (set) when heated, "binding" other ingredients together. (e.g., in meatballs, meatloaf, and some pastries).
2. **THICKENING:** Egg yolk, in particular, is used to thicken sauces and custards. The protein coagulates and creates a thick, smooth texture. (e.g., in custard, mayonnaise).
3. **LEAVENING (AERATION):** Egg whites, when whisked, can trap air and increase in volume by 6-8 times. This foam, when folded into a mixture, acts as a raising agent. (e.g., in foam cakes like Angel food, meringues, and soufflés).
4. **EMULSIFYING:** The *lecithin* in the egg yolk allows two liquids that normally repel each other (like oil and water) to mix without separating. (e.g., in mayonnaise, hollandaise sauce, salad dressing).
5. **GLAZING:** A beaten egg (or egg wash) brushed onto baked goods gives them a shiny, brown finish. (e.g., on meat pies, bread, biscuits).
6. **CLARIFYING:** Egg whites are used to "clear" soups and stocks. When added, they coagulate and trap impurities, which can then be skimmed off.
7. **AS A MAIN DISH:** Boiled, fried, poached, or scrambled.

4: TEST FOR FRESHNESS

As an egg ages, its air cell grows, and the white becomes thinner.

1. **THE WATER TEST (FLOAT TEST):**
 - Place the egg in a bowl of water.
 - **FRESH EGG:** Sinks to the bottom and lies flat on its side.
 - **SLIGHTLY OLDER EGG (STILL GOOD):** Sinks, but the blunt end floats upwards.
 - **STALE/BAD EGG:** Floats on the surface. (The air cell is so large it buoys the egg). **DO NOT USE.**
2. **THE BREAK TEST:**
 - Crack the egg onto a flat plate.

- **FRESH EGG:** The yolk is high, domed, and centred. The thick albumen stands up firmly around it, separate from the thin white.
- **STALE EGG:** The yolk is flat. The albumen is thin and watery, spreading out across the plate.

5: METHODS OF COOKING EGGS AND EGG DISHES

- **THE KEY TO EGG COOKERY:** Cook *gently* (low-to-medium heat). High heat makes the proteins coagulate too quickly, resulting in a tough, rubbery, and dry product.
- **METHODS:**
 1. **BOILING:**
 - **Hard-Boiled:** Simmered in water for 10-12 minutes. The yolk and white are solid.
 - **Soft-Boiled:** Simmered for 3-5 minutes. The white is set, but the yolk is runny.
 2. **FRYING (SUNNY-SIDE UP, OVER EASY):** Fried in a small amount of fat.
 3. **POACHING:** Cracked into a pan of simmering (not boiling) water.
 4. **SCRAMBLING:** Whisked (often with milk) and cooked gently in a pan while stirring.
 5. **BAKING:** Baked in a dish (e.g., "shakshuka" or as part of a quiche).
 6. **OMELETTE:** Whisked eggs cooked in a pan without stirring, then folded.
- **EGG DISHES:**
 - Scrambled eggs
 - Omelettes (plain or filled)
 - Fried egg (for bread, noodles, or rice)
 - Egg stew / Egg sauce
 - Custard
 - Mayonnaise
 - Meringue
 - Scotch eggs

EVALUATION

1. Draw a cross-section of an egg and label the four main parts (shell, albumen, yolk, air cell).
2. What is the "float test," and what does it mean if an egg floats in water?
3. List and explain three different culinary *functions* of eggs (e.g., binding, thickening...).
4. What component in egg yolk makes it a good *emulsifier*?
5. What is the "golden rule" of egg cookery (regarding heat), and why is it important?

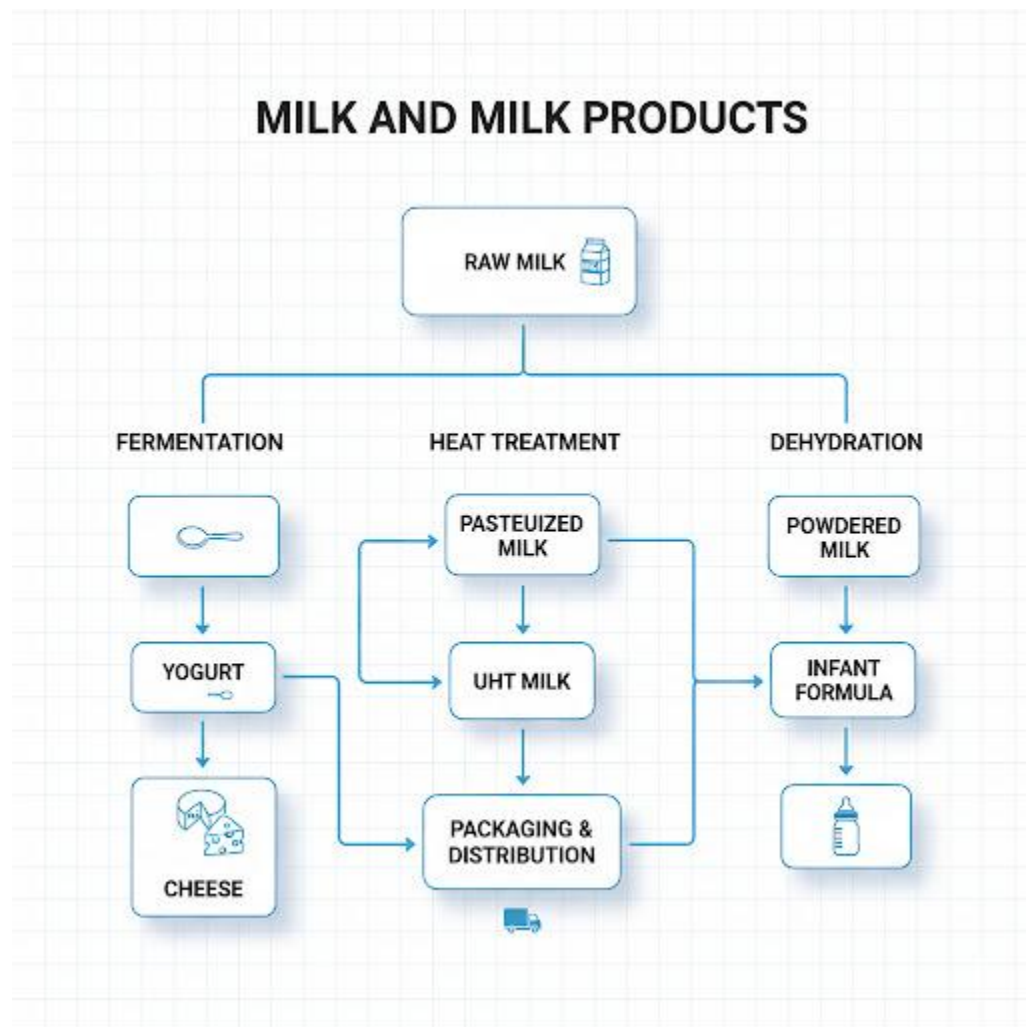
ASSIGNMENT

1. Perform the "float test" at home with an egg. Then, perform the "break test." Write a one-paragraph report on your findings, describing the freshness of your egg based on your observations.
2. Write a 400-word essay on the nutritional benefits of eggs. Also, discuss the (now outdated) controversy surrounding eggs and dietary cholesterol.
3. What is "mayonnaise"? Research its ingredients and explain *why* the egg yolk is the most important ingredient.
4. Develop a recipe for a "Nigerian-style" omelette, listing the ingredients (e.g., onion, pepper, tomato) and the step-by-step cooking method.
5. What is a "meringue"? What part of the egg is it made from, and what is the scientific principle behind its formation?

WEEK 4: MILK AND MILK PRODUCTS

TOPIC: MILK AND MILK PRODUCTS

INTRODUCTION: Milk is the first food of all mammals, providing a complete source of nutrition. In human diets, "milk" typically refers to cow's milk, but it can also come from goats, sheep, and other animals. It is a highly nutritious and versatile food, but also highly perishable.



1: MEANING, TYPES, AND NUTRITIVE VALUE

- **MEANING:** Milk is a white, nutrient-rich liquid secreted by the mammary glands of mammals.
- **NUTRITIVE VALUE:**

- **PROTEIN:** Contains high-quality protein (Casein and Whey).
- **CARBOHYDRATE:** Contains a milk sugar called **Lactose**.
- **FAT:** The fat content varies (e.g., whole, skimmed). It is a source of fat-soluble vitamins.
- **MINERALS:** The *best* food source of **Calcium**, which is essential for strong bones and teeth. Also rich in Phosphorus.
- **VITAMINS:** Rich in **Vitamin A** (in the fat), **Vitamin D** (often fortified), and **Riboflavin (B2)**.
- **TYPES OF MILK (BASED ON PROCESSING):**
 1. **FRESH MILK (LIQUID):**
 - **Pasteurized:** Heated to kill harmful bacteria, extending its short shelf life. Must be refrigerated.
 - **Homogenized:** A process that breaks down fat globules to prevent a cream layer from forming. Most pasteurized milk is also homogenized.
 - **Whole Milk:** Full fat (3.5-4%).
 - **Low-Fat / Skimmed Milk:** Fat has been (partially or fully) removed.
 2. **EVAPORATED MILK:** About 60% of the water is removed. It is a thick, concentrated, sterilized milk sold in cans.
 3. **SWEETENED CONDENSED MILK:** Similar to evaporated milk, but a large amount of sugar is added. The sugar acts as a preservative.
 4. **POWDERED (DRY) MILK:** All water has been removed, creating a powder with a very long shelf life.
 5. **UHT (ULTRA-HIGH TEMPERATURE) MILK:** Milk is heated to a very high temperature for a few seconds, sterilizing it completely. It can be stored (unopened) at room temperature for months.

2: METHODS OF PRESERVING MILK

Milk is highly perishable because its rich nutrient content and high water content make it a perfect breeding ground for bacteria.

1. **PASTEURIZATION:** Heating milk to 72°C for 15 seconds to kill pathogenic (disease-causing) bacteria. It *does not* kill all bacteria, so the milk *must* be kept refrigerated and has a limited shelf life.
2. **STERILIZATION:** Heating milk to a higher temperature (100°C+) for a longer time. This kills *all* bacteria. It gives the milk a longer shelf life but can alter the taste and destroy some vitamins.
3. **UHT (ULTRA-HIGH TEMPERATURE) PROCESSING:** Heating to 135-150°C for 1-2 seconds. This sterilizes the milk, which is then packaged in sterile, airtight containers.
4. **EVAPORATION:** Removing water (as in evaporated milk).
5. **DEHYDRATION (DRYING):** Removing all water (as in powdered milk).
6. **FERMENTATION:** Using "good" bacteria to ferment the lactose into lactic acid. The acid prevents "bad" bacteria from growing. (e.g., Yogurt).
7. **ADDITION OF PRESERVATIVES:** Adding sugar (as in condensed milk).

3: MILK PRODUCTS

These are foods produced from milk.

1. **YOGURT:** Milk fermented by specific "live" bacterial cultures. It is a probiotic.
2. **CHEESE:** Produced by coagulating the protein (casein) in milk. There are thousands of varieties (e.g., Cheddar, Mozzarella, *Wara* - Nigerian cheese).
3. **BUTTER:** Made by churning cream (the high-fat part of milk) to separate the fat (butterfat) from the liquid (buttermilk).
4. **CREAM:** The high-fat layer that is skimmed off the top of milk.
5. **ICE CREAM:** A frozen dessert made from cream, milk, sugar, and flavourings.
6. **"KINDIRMO" (NUNU):** A fermented milk product, similar to yogurt, popular in Northern Nigeria.

4: EXAMPLES OF MILK DISHES

- **DRINKS:** Milkshakes, Smoothies.
- **BREAKFAST:** Pap (Ogi) with milk, Cereal with milk, Custard.
- **DESSERTS:** Ice cream, Yogurt, Milk pudding (e.g., rice pudding).

- **BAKING:** Used as a liquid in cakes, bread, and pastries.
- **SAVOURY:** Used as a base for white sauces (Béchamel sauce), in soups, or in cheese-based dishes.

EVALUATION

1. Why is milk considered a "complete food," and what is its most important mineral contribution to the diet?
2. What is the difference between "Pasteurization" and "UHT" processing? Which one results in a longer shelf life at room temperature?
3. What is "Lactose"? What is "Lactose Intolerance"?
4. Explain the difference between Evaporated Milk and Sweetened Condensed Milk.
5. List three milk products (e.g., yogurt, cheese) and one milk-based dish (e.g., custard).

ASSIGNMENT

1. Milk is a "perfect" food for bacteria. Write a 300-word report explaining the three main reasons why milk spoils so quickly.
2. Research the process of "Homogenization." What is it, and why is it done? (Hint: it's not for safety).
3. What is "Wara"? Research how this local Nigerian cheese is made and what is used to coagulate the milk.
4. Read the label on three different types of milk in a shop (e.g., powdered, evaporated, UHT). Create a table comparing their nutritional information, ingredients, and storage instructions.
5. What is "Casein" and what is "Whey"? These are the two proteins in milk. Research their uses in the food industry (e.s., protein powder).

WEEK 5: MIDTERM TEST

TOPIC: ASSESSMENT OF WEEKS 1–4

INTRODUCTION: This week is dedicated to a formal assessment of all concepts covered in the first part of the term. The test will evaluate your understanding of meat, poultry, eggs, and milk.

SCOPE:

- **WEEK 1: MEAT COOKERY:** Types, structure (muscle, fat, collagen, elastin), nutritive value, factors in choice, cooking methods (dry vs. moist).
- **WEEK 2: POULTRY:** Types (broiler, hen), structure (white/dark meat), nutritive value, preparation (trussing, safety), cooking methods.
- **WEEK 3: EGGS:** Structure (yolk, albumen, air cell), nutritive value, culinary uses (binding, leavening, etc.), freshness tests.
- **WEEK 4: MILK:** Types (pasteurized, UHT, evaporated, powdered), nutritive value (calcium), preservation methods, milk products (yogurt, cheese).

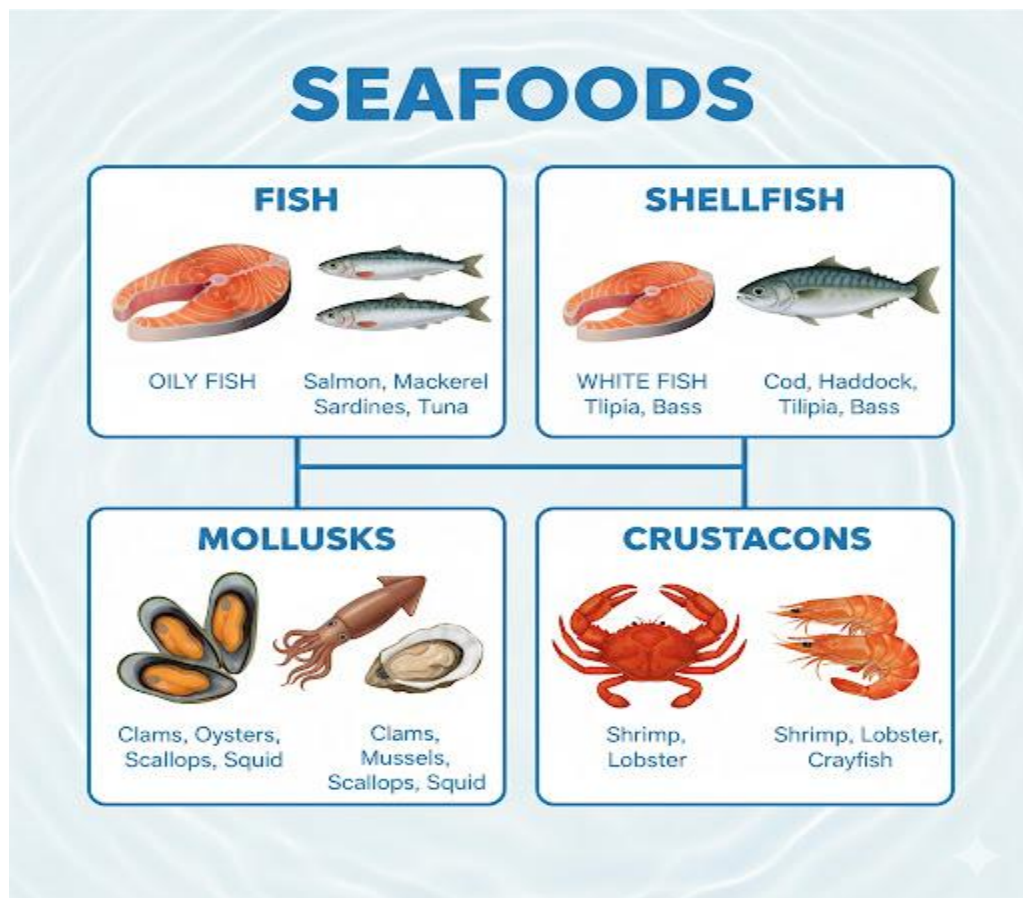
FORMAT:

- **SECTION A: MULTIPLE CHOICE QUESTIONS (20 MARKS)** - Testing definitions and key facts.
- **SECTION B: SHORT ANSWER QUESTIONS (30 MARKS)** - Testing explanations (e.g., "Explain the float test for eggs," "Differentiate between stewing and braising").
- **SECTION C: ESSAY/LONG ANSWER QUESTIONS (50 MARKS)** - (e.g., "Discuss the structure of meat and how it determines the cooking method," "Explain the five main culinary uses of eggs with examples").

WEEK 6: SEA FOODS

TOPIC: SEA FOODS (FISH AND SHELLFISH)

INTRODUCTION: "Sea food" is a broad term for all edible aquatic animals harvested from the sea or from freshwater. This category includes fish and shellfish, which are important sources of protein and unique nutrients like omega-3 fatty acids and iodine.



1: MEANING AND CLASSIFICATION OF SEA FOODS

- **MEANING:** Sea foods (or "seafood") are aquatic animals used as food.
- **CLASSIFICATION:**
 1. **FISH (FINFISH):**
 - These are vertebrates (have a backbone).

- **Classification by Habitat:**
 - **Saltwater Fish:** (e.g., Mackerel, Tuna, Sardine, Cod, Croaker).
 - **Freshwater Fish:** (e.g., Tilapia, Catfish).
- **Classification by Fat Content (Important for Cookery):**
 - **OILY FISH (FAT FISH):** Contain 10-20% fat distributed throughout the flesh. (e.g., Mackerel, Salmon, Tuna, Sardine, Herring). Rich in Omega-3.
 - **WHITE FISH (LEAN FISH):** Contain <5% fat, stored mainly in the liver (which is removed). (e.g., Tilapia, Cod, Croaker, Whiting, Flounder).

2. SHELLFISH:

- These are invertebrates (have no backbone).
- **Classification by Shell Type:**
 - **MOLLUSKS:** Have a soft body and a hard, hinged shell. (e.g., Oysters, Mussels, Clams, Scallops, Periwinkles - *Isam*).
 - **CRUSTACEANS:** Have a segmented body and a hard, jointed, non-hinged shell (exoskeleton). (e.g., Prawns/Shrimp, Crabs, Lobsters, Crayfish).

2: NUTRITIVE VALUE

- **PROTEIN:** Excellent source of high-biological-value protein. The connective tissue in fish is weaker than in meat, making it naturally tender and quick to cook.
- **FAT (OMEGA-3 FATTY ACIDS):** Oily fish are the best source of **Omega-3s (EPA & DHA)**, which are crucial for brain health, heart health, and reducing inflammation.
- **MINERALS:**
 - **IODINE:** Sea foods are the best natural source. Iodine is essential for thyroid function (preventing goiter).
 - **ZINC & SELENIUM:** Important antioxidants.
 - **CALCIUM:** Canned fish with soft, edible bones (like sardines) are an excellent source of calcium.

- **VITAMINS:**

- **Vitamin D & A:** Found in high amounts in oily fish (especially the liver).
- **B-Complex Vitamins.**

3: FACTORS TO CONSIDER WHEN CHOOSING SEA FOODS

- Sea food is highly perishable and can cause severe food poisoning if not fresh.
- **CHOOSING FRESH FISH (WHOLE):**
 - **EYES:** Should be bright, clear, and bulging (not cloudy or sunken).
 - **GILLS:** Should be bright red (not brown or slimy).
 - **FLESH:** Should be firm and spring back when pressed (no fingerprint left).
 - **SMELL:** Should smell "fresh and briny" (like the sea), *not* "fishy" or like ammonia.
 - **SCALES:** Should be shiny and cling tightly to the skin.
- **CHOOSING FRESH SHELLFISH:**
 - **PRAWNS/SHRIMP:** Should be firm. A "mushy" texture means it's old.
 - **CRABS/LOBSTERS:** Should be heavy for their size. If bought live, they should be active.
 - **MOLLUSKS (CLAMS, OYSTERS):** If bought live, the shell should be *tightly closed*. If the shell is open, tap it. If it doesn't close, it is dead. **Discard it.**

4: METHODS OF COOKING FISHES

- **THE KEY TO FISH COOKERY:** Fish cooks *very quickly* and is easily *overcooked*. The goal is to cook it until the flesh just turns opaque (no longer translucent) and "flakes" easily with a fork.
- **DRY-HEAT METHODS:**
 - **FRYING (SHALLOW, DEEP):** Very common (e.g., fried fish for stew, fish and chips).
 - **GRILLING/BARBECUING:** Excellent for oily fish (e.g., grilled mackerel) or whole fish.
 - **BAKING/ROASTING:** Good for whole fish or fish fillets.

- **SMOKING:** A form of preservation and cooking (e.g., smoked fish).
- **MOIST-HEAT METHODS:**
 - **POACHING:** Gently simmering the fish in liquid (water, stock, milk). Good for delicate white fish.
 - **STEAMING:** Cooking the fish over boiling water. Preserves nutrients and flavour.
 - **STEWING:** Adding fish (usually pre-fried or smoked) to a stew or soup base (e.g., Nigerian fish stew, fish pepper soup).

EVALUATION

1. What is the main difference between "Fish" and "Shellfish"?
2. What is the main nutritional difference between "Oily Fish" and "White Fish"? Give one example of each.
3. List three specific signs to look for when buying a *fresh whole fish*.
4. Why must you discard a live mussel or oyster if its shell is open and does not close when tapped?
5. What is the "golden rule" of cooking fish? (Hint: it relates to time/doneness).

ASSIGNMENT

1. Write a 400-word report on the health benefits of Omega-3 fatty acids, explaining *why* doctors recommend eating oily fish.
2. What is "Iodine," and what is "Goiter"? Research and explain the link between sea food consumption and the prevention of this disease.
3. Go to a market and identify five different types of sea foods sold (e.g., Mackerel, Tilapia, Prawns, Periwinkles, Smoked fish). For each, list its local name (if any) and its price.
4. Develop a step-by-step recipe for "Fish Pepper Soup," listing all ingredients and the method.
5. Research the process of "smoking" fish. How does it preserve the fish, and what are the two main methods (hot vs. cold)?

WEEK 7: MIDTERM BREAK



WEEK 8: FOOD ADDITIVES

TOPIC: FOOD ADDITIVES, FLAVOURINGS, HERBS, AND SPICES

INTRODUCTION: Food additives are substances added to food to maintain or improve its safety, freshness, taste, texture, or appearance. These can range from natural spices and herbs used for centuries to commercially produced chemical compounds used in modern food processing.



1: MEANING AND TYPES OF FOOD ADDITIVES

- **MEANING:** A food additive is any substance intentionally added to a food product in small quantities to achieve a specific technical function.
- **FUNCTIONS (TYPES):**
 1. **PRESERVATIVES:** Extend shelf-life by preventing spoilage by microbes (bacteria, yeast, mold) or oxidation.
 - **Natural:** Salt (for fish/meat), Sugar (for jam), Vinegar.
 - **Chemical:** Benzoates, Sorbates.

2. **FLAVOURINGS & FLAVOUR ENHANCERS:** Add or intensify taste and aroma.
 - **Flavourings:** (e.g., Vanilla essence, artificial banana flavour).
 - **Enhancers:** (e.g., Monosodium Glutamate - MSG, in stock cubes).
3. **COLOURINGS:** Add or restore colour to food to make it visually appealing. (e.g., Tartrazine, Sunset Yellow, natural colours like Riboflavin).
4. **SWEETENERS:** Provide sweetness.
 - **Nutritive:** Sugar, Glucose.
 - **Non-Nutritive (Artificial):** Saccharin, Aspartame (for "diet" drinks).
5. **THICKENERS, STABILIZERS & EMULSIFIERS:** Control the texture and consistency. (e.g., Pectin, Gelatin, Lecithin).

2: CLASSIFICATION (LOCAL VS. FOREIGN)

This classification focuses on flavourings, which are a *type* of additive.

- **LOCAL FLAVOURINGS (NIGERIAN):**
 - **HERBS (LEAVES):**
 - **Scent Leaf (Efirin):** Pungent, minty/clove flavour. Used in pepper soup, stews.
 - **Curry Leaf (Eweta):** Aromatic, used in stews and sauces.
 - **Bitter Leaf (Onugbu):** Has a strong bitter taste, must be washed. Used in Bitterleaf soup.
 - **Uziza Leaf:** Spicy and aromatic, used in soups.
 - **SPICES (SEEDS, BARKS, ROOTS):**
 - **Uziza Seed:** Spicy, aromatic.
 - **Ehuru (African Nutmeg):** Aromatic, smoky.
 - **Iru / Ogiri / Dawadawa:** Fermented locust beans, pungent aroma, adds "umami" flavour.
 - **Aidan Fruit:** A strong, aromatic pod.
 - **Ginger, Garlic, Onion:** These are used locally and globally.
- **FOREIGN FLAVOURINGS (GLOBAL):**

- **HERBS (LEAVES):**
 - **Thyme:** Earthy, slightly minty. Used for meat, stews.
 - **Rosemary:** Strong, pine-like. Used for roasts (lamb, chicken).
 - **Bay Leaf:** Adds a subtle background flavour to stews and Jollof rice.
 - **Parsley:** Mild, fresh flavour, often used as a garnish.
 - **Basil:** Sweet, peppery. (Used in Italian food).
- **SPICES (SEEDS, BARKS, ROOTS):**
 - **Black Pepper:** Pungent, hot.
 - **Curry Powder:** A *blend* of spices (e.g., turmeric, cumin, coriander).
 - **Thyme (Dry):** Often sold as a spice.
 - **Nutmeg, Cinnamon, Cloves:** "Sweet" spices used in baking and drinks (zobo).

3: USES AND PRESENTATION ON ALBUM/CARDBOARD

- **USES OF HERBS AND SPICES:**
 1. **ADD FLAVOUR AND AROMA:** The primary use. They make food interesting.
 2. **PRESERVATION:** Some, like cloves and salt, have anti-microbial properties.
 3. **APPETITE STIMULATION:** The aroma can stimulate the appetite.
 4. **NUTRITION & MEDICINE:** Many, like ginger and garlic, have well-known medicinal properties (anti-inflammatory, anti-bacterial).
 5. **GARNISHING:** Fresh herbs (like parsley or scent leaf) make a dish look appealing.
- **PRACTICAL ACTIVITY: HERB & SPICE ALBUM:**
 - **OBJECTIVE:** To create a visual reference for identifying local and foreign herbs and spices.
 - **METHOD:**
 1. Students will collect small, *dried* samples of various herbs and spices.
 2. Arrange them neatly on a large piece of cardboard or in a scrapbook ("album").
 3. Samples can be placed in small, sealed transparent bags.

4. Next to each sample, create a label with:
 - **Name (English):** e.g., "African Nutmeg"
 - **Local Name (Yoruba/Igbo/Hausa):** e.g., "Ehuru"
 - **Classification:** (Local / Foreign)
 - **Common Use:** e.g., "Used in pepper soup and stews"

EVALUATION

1. What is a "food additive"?
2. List the five main functions/types of food additives.
3. What is the primary difference between an "herb" and a "spice"?
4. Name two local (Nigerian) herbs and two local (Nigerian) spices.
5. What is the purpose of a flavour *enhancer* like MSG (stock cubes)?

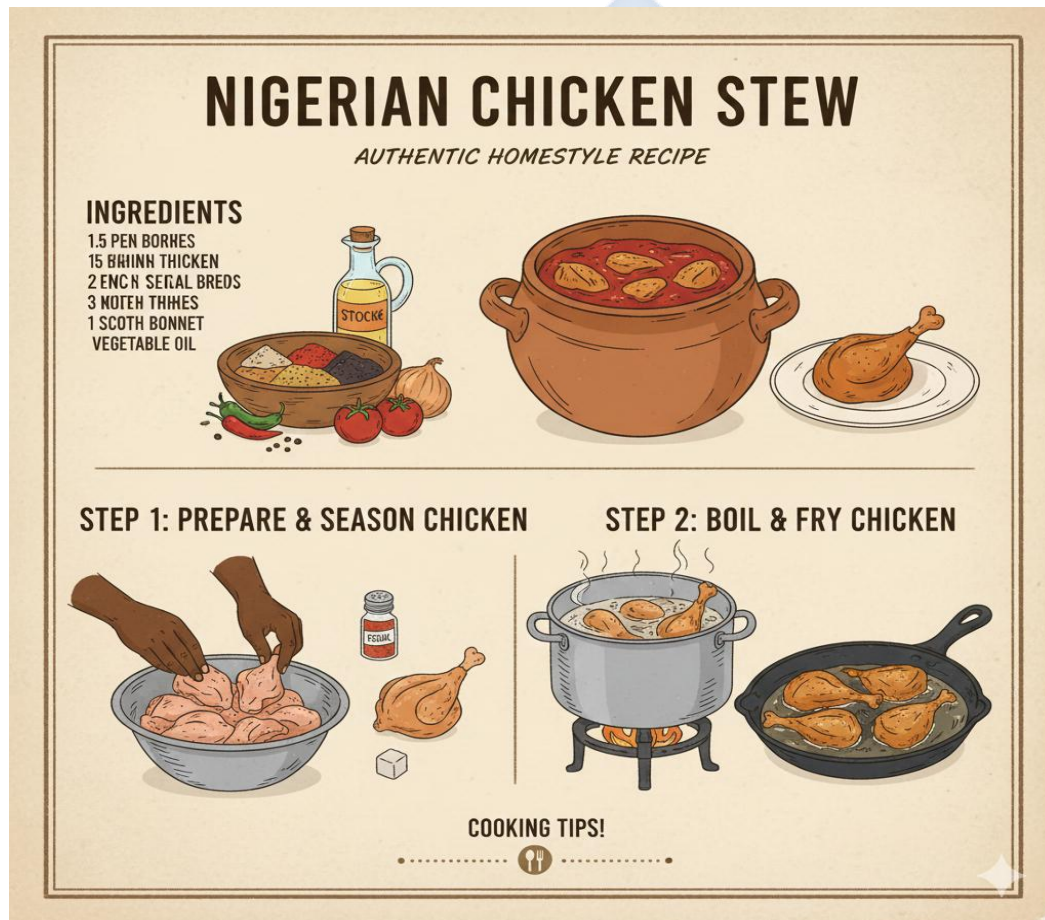
ASSIGNMENT

1. Go to your kitchen at home. Find five different food items that have an "Ingredients" label (e.g., stock cubes, tinned milk, a "diet" drink, bread).
2. Read the labels and identify *all* the food additives listed.
3. Create a table with three columns: (1) Product Name, (2) Additive Name, (3) Purpose of the Additive (e.g., "Preservative," "Colouring," "Sweetener").
4. What is the difference between "Curry Leaf" (local) and "Curry Powder" (foreign)? Research and explain.
5. Start your Herb & Spice Album. Collect, press (if fresh), and label at least 5 different local herbs/spices and 3 different foreign herbs/spices. (This will be your submission for the practical).

WEEK 9: PRACTICAL ON MEAT, POULTRY, MILK, SEA FOODS, AND FOOD ADDITIVES

TOPIC: PREPARATION AND SERVING OF DISHES

INTRODUCTION: This week is purely practical. The objective is to apply the theoretical knowledge gained in Weeks 1, 2, 4, 6, and 8. Students will plan, prepare, cook, and serve a dish that incorporates one or more of the main protein foods (meat, poultry, sea food) and demonstrates the correct use of herbs, spices, and additives.



1: PRACTICAL OBJECTIVES

1. To demonstrate competence in handling high-risk protein foods (meat, poultry, fish) safely.

2. To select and apply the *correct* cooking method for a specific cut of meat or type of poultry/fish.
3. To use local and/or foreign additives (herbs, spices, flavourings) to enhance the flavour and appearance of a dish.
4. To work efficiently and hygienically in a kitchen environment.
5. To present the final dish in an appealing and appetizing manner.

2: SUGGESTED DISHES (CHOOSE ONE)

The choice of dish allows students to integrate different topics:

- **1. NIGERIAN CHICKEN/GOAT MEAT STEW:**
 - **Topics Covered:** Meat/Poultry (tough cut), Moist-Heat Cookery (stewing), Additives (curry, thyme, bay leaf, stock cubes, local spices).
 - **Skills:** Disjointing poultry, boiling, frying, and stewing.
- **2. GRILLED MACKEREL (TITUS) WITH PEPPER SAUCE:**
 - **Topics Covered:** Sea Food (oily fish), Dry-Heat Cookery (grilling), Additives (herbs/spices for the marinade/sauce).
 - **Skills:** Cleaning/gutting fish, marinating, grilling, sauce-making.
- **3. CHICKEN/BEEF SUYA (MINI):**
 - **Topics Covered:** Meat/Poultry (tender cut), Dry-Heat Cookery (grilling), Additives (Yaji - a local spice blend).
 - **Skills:** Slicing meat thinly, marinating, skewering, grilling.
- **4. YAM AND FISH PEPPER SOUP:**
 - **Topics Covered:** Sea Food (fish - e.g., catfish), Starchy Roots (from Week 10), Additives (local pepper soup spice mix, scent leaf).
 - **Skills:** Preparing fish, using local spice blends, moist-heat cookery.
- **5. MILK-BASED DISH (E.G., CUSTARD WITH FRUIT OR A SIMPLE "WAPA")**
 - **Topics Covered:** Milk, Egg (for custard), Thickening (using eggs or cornflour), Additives (vanilla essence, nutmeg).
 - **Skills:** Thickening, pasteurization (if making wara), gentle heat.



3: PLANNING AND SAFETY

1. **PLANNING:** Students must submit a "Plan of Work" (recipe, ingredients list, equipment list) *before* the practical day.
2. **HYGIENE:**
 - Clean aprons, hairnets, and short fingernails are mandatory.
 - Hand-washing is the first and last step.
 - Separate chopping boards *must* be used for raw protein and other ingredients (vegetables).
 - All surfaces must be sanitized after use.
3. **SAFETY:**
 - Be cautious with hot oil (frying).
 - Be careful with knives.

- Cook all meat, poultry, and fish *thoroughly* to the correct internal temperature.

EVALUATION (PRACTICAL ASSESSMENT)

Students will be graded on:

1. **PLANNING (10%):** Quality of the submitted plan.
2. **HYGIENE & SAFETY (20%):** How clean and safe they worked.
3. **SKILL & METHOD (40%):** Correct application of cooking skills (e.g., did they stew the tough meat? Was the fish overcooked?).
4. **FLAVOUR & TEXTURE (20%):** The taste and palatability of the final dish.
5. **PRESENTATION (10%):** How the final dish is served.

ASSIGNMENT (POST-PRACTICAL)

1. Write a detailed "Post-Practical Report" on the dish you cooked.
2. The report must include: (a) The recipe, (b) The steps you took, (c) One *challenge* you faced (e.g., "The fish broke apart"), and (d) One thing you learned or would do differently next time.
3. Why was the *specific* herb/spice you used (e.g., scent leaf, thyme) a good choice for your dish?
4. Explain the main food safety risk of the *raw ingredient* you handled (e.g., Salmonella in chicken, bacteria in fish).
5. Cost your final dish. List all the ingredients you used and their approximate market price to find the "cost per serving."

WEEK 10: STARCHY ROOTS AND PLANTAIN

TOPIC: STARCHY ROOTS AND PLANTAIN

INTRODUCTION: Starchy roots, tubers, and corms, along with plantain, form the "staple" foods for millions of people in West Africa and other tropical regions. They are the primary source of dietary energy (calories) and a key part of the cultural diet.



1: STARCHY ROOTS (ROOTS AND TUBERS)

- **MEANING:** These are underground plant organs that are swollen with stored starch.

- **True Roots:** (e.g., Cassava, Sweet Potato).
- **Tubers (Swollen Stems):** (e.g., Yam, Potato).
- **Corms (Swollen Stems):** (e.g., Cocoyam).
- **TYPES:**
 1. **CASSAVA (MANIHOT ESCULENTA):**
 - A woody shrub with a starchy root. A major staple.
 - **Varieties:** Sweet and Bitter. Bitter cassava contains *cyanide* and **must** be processed (e.g., by fermenting, boiling) to be safe.
 - **Products:** Gari, Fufu, Lafun, Starch, Tapioca.
 2. **YAM (DIOSCOREA SPP.):**
 - A staple in West Africa (the "Yam Belt").
 - A large, starchy tuber. Highly valued culturally.
 - **Products:** Pounded Yam, Boiled Yam, Fried Yam, Amala (from yam flour).
 3. **COCOYAM (XANTHOSOMA SAGITTIFOLIUM):**
 - A "corm." Often used as a thickener for soups or boiled.
 - **Products:** Boiled, used in *Achu* or to thicken *Oha* soup.
 4. **SWEET POTATO:**
 - A sweet-tasting root. Can be white, yellow, or orange-fleshed.
 - **Products:** Boiled, Fried, Roasted.

2: NUTRITIVE VALUE OF STARCHY ROOTS

- **CARBOHYDRATES (STARCH):** This is their primary function. They are an excellent, high-energy, low-cost source of calories.
- **FIBRE:** Good source of dietary fibre, which aids digestion.
- **VITAMINS:**
 - Generally a good source of **Vitamin C** (especially yam).
 - **Vitamin A:** *Orange-fleshed* sweet potatoes are an extremely rich source of Vitamin A (Beta-carotene).
- **MINERALS:** Contain potassium and magnesium.

- **PROTEIN & FAT:** They are *very poor* sources of protein and fat. This is why they must be eaten with protein-rich stews (e.g., meat, fish) to form a balanced diet.

3: PLANTAIN (MUSA PARADISIACA)

- **MEANING:** A type of cooking banana. It is a staple fruit that is eaten *cooked*. It is classified based on its ripeness, which changes its flavour, texture, and use.

- **TYPES AND VALUE:**

1. **UNRIPE PLANTAIN (GREEN):**

- **Value:** Very high in starch and fibre. Has a firm texture and a bland, starchy taste (like a potato).
- **Uses:** Boiled (eaten with stew), Fried (to make chips), Roasted (Boli), or dried and milled into *Plantain Flour* (Amala).

2. **RIPE PLANTAIN (YELLOW):**

- **Value:** The starches have converted to *sugars*. It is soft, sweet, and aromatic.
- **Uses:** Frying (to make *Dodo*), Boiling (to make *Kekefia* - a pottage), Roasting (Boli - becomes soft and caramelized).

3. **OVER-RIPE PLANTAIN (BLACK):**

- **Value:** Very soft and extremely sweet.
- **Uses:** Mashed and fried (to make *Mosa* or plantain puff-puff).

4: DISHES FROM PLANTAIN AND STARCHY ROOTS

- **CASSAVA:** Gari (Eba), Fufu, Cassava fufu (Akpu), Abacha (African salad).
- **YAM:** Pounded Yam (Iyan), Amala (from Yam flour), Yam Pottage, Boiled/Fried Yam.
- **PLANTAIN:** Dodo (fried ripe), Plantain chips (fried unripe), Boli (roasted), Plantain pottage.

EVALUATION

1. What is the primary nutritional function of starchy roots and plantain in the diet?
2. What major nutrient are starchy roots *lacking*?

3. What is the key chemical and culinary difference between unripe (green) plantain and ripe (yellow) plantain?
4. Why is it dangerous to eat "Bitter Cassava" without processing it, and what products are made from processed cassava?
5. Name one dish made from Yam and one dish made from ripe Plantain.

ASSIGNMENT

1. A diet consisting *only* of Gari (from cassava) and water leads to a severe protein-deficiency disease. Research and name this disease (it begins with 'K'), and explain *why* the diet causes it.
2. What is "Vitamin A deficiency," and why is it a problem? Explain how *orange-fleshed sweet potato* is being used to solve this.
3. Write a detailed, step-by-step recipe for your favorite plantain dish (e.g., Dodo, Boli, or Plantain Pottage).
4. "Pounded Yam" (Iyan) is a traditional dish. What is the modern, "convenience" product that is used to make it today?
5. Go to a market. Find and list the prices for 1 small tuber of Yam, 1 "hand" of Plantain, and 1 "mudu" (cup) of Gari. Which do you think provides the cheapest calories?

WEEK 11: REVISION

TOPIC: GENERAL REVIEW OF ALL TOPICS

INTRODUCTION: This week is dedicated to a comprehensive review of all topics covered in SS2 First Term. We will consolidate our knowledge and prepare for the final examinations, focusing on the connections between topics.



REVISION SCOPE:

- **PROTEIN FOODS:**

- **MEAT:** Structure (collagen), types (red/white), cooking (dry/moist).
- **POULTRY:** Structure (dark/white), types (broiler/hen), safety (Salmonella).
- **EGGS:** Structure, freshness tests, 7 culinary uses.
- **MILK:** Preservation (pasteurization/UHT), products (yogurt/cheese), nutrients (calcium).
- **SEA FOODS:** Classification (oily/white, mollusk/crustacean), freshness signs, nutrients (Omega-3, Iodine).
- **FLAVOUR & ENERGY:**
 - **ADDITIVES:** Herbs (local/foreign) vs. Spices (local/foreign), types (preservatives, colouring, etc.).
 - **STARCHY ROOTS & PLANTAIN:** Types (yam, cassava), products (gari, fufu), plantain (ripe/unripe), nutritive value (starch).
- **PRACTICALS:**
 - Safety and hygiene.
 - Matching cooking methods to ingredients.
 - Planning and presentation.

REVISION STRATEGY:

1. **Review all notes.**
2. **Answer all "Evaluation" questions** from each week.
3. **Create "Mind Maps"** connecting topics (e.g., a mind map for "Protein" showing all 5 types).
4. **Practice essay questions:**
 - "Compare and contrast the nutritive value of Meat and Fish."
 - "Discuss the 7 culinary uses of eggs, giving a specific dish for each."
 - "Explain the key differences between dry-heat and moist-heat cookery, giving examples of cuts of meat suitable for each."

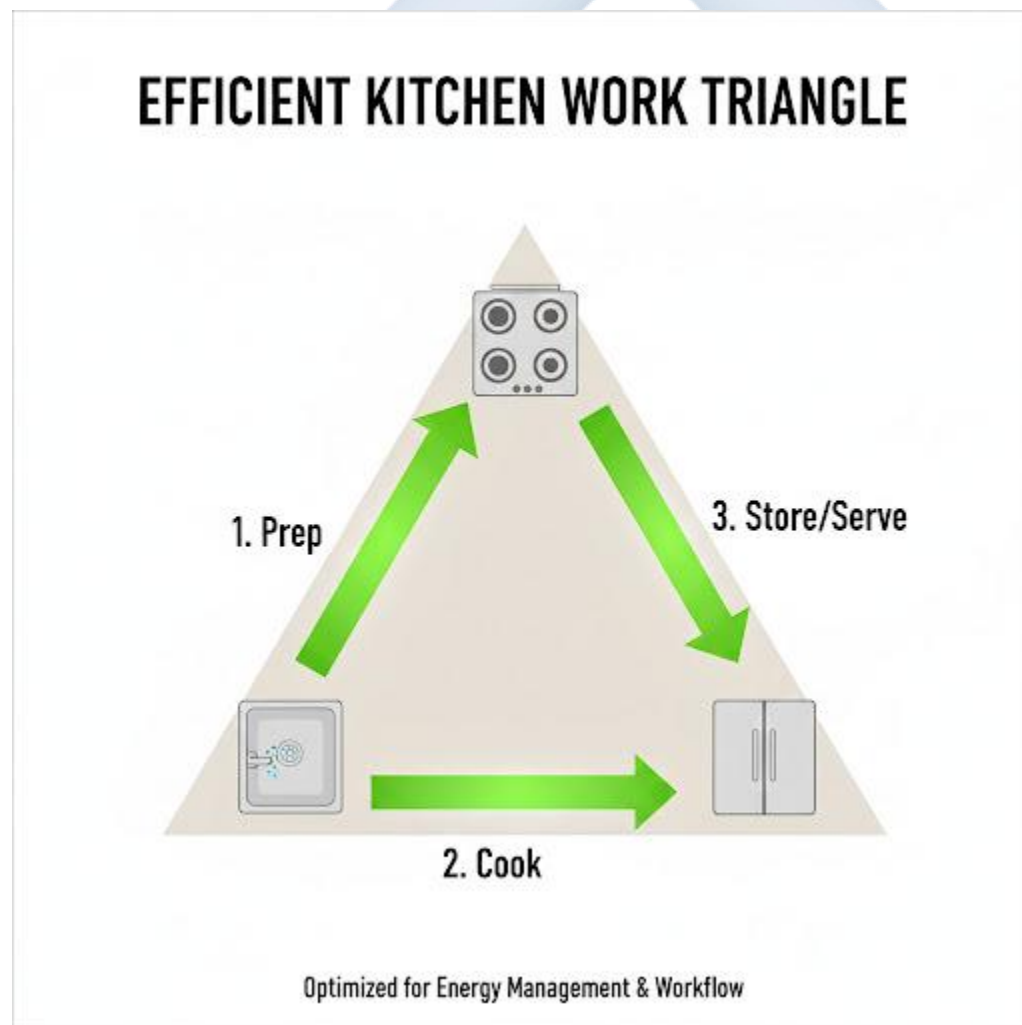
SS2 FOOD AND NUTRITION LESSON NOTES

SECOND TERM

WEEK 1: ENERGY MANAGEMENT IN FOOD PREPARATION

TOPIC: ENERGY MANAGEMENT IN FOOD PREPARATION

INTRODUCTION: Energy is a critical resource in the home, and its management is a key aspect of home economics. In food preparation, "energy" refers not only to the fuel used (electricity, gas, kerosene) but also to the human effort (time and physical work) required. Effective energy management involves using all these resources wisely to achieve maximum results with minimum waste.



1: MEANING AND ADVANTAGES OF ENERGY MANAGEMENT

- **MEANING OF ENERGY MANAGEMENT:** This is the conscious and careful planning, control, and use of resources (human and non-human) to complete tasks efficiently, effectively, and with minimal waste.
 - **HUMAN ENERGY:** Time, physical effort, mental effort, skill.
 - **NON-HUMAN ENERGY (FUEL):** Electricity, cooking gas, kerosene, charcoal, firewood.
- **ADVANTAGES OF ENERGY MANAGEMENT:**
 1. **SAVES TIME:** A well-managed plan allows tasks to be completed in the shortest possible period, freeing up time for other activities (leisure, work, family).
 2. **SAVES MONEY:** Efficient use of fuel (e.g., using a lid on a pot, cooking multiple items at once) directly reduces household bills (gas, electricity). It also prevents waste of food ingredients.
 3. **REDUCES FATIGUE:** This is the primary human benefit. By planning work and using energy-saving techniques, you end the task with less physical and mental exhaustion.
 4. **INCREASES EFFICIENCY (PRODUCTIVITY):** More work is accomplished with the same amount of effort.
 5. **IMPROVES SAFETY:** A well-planned, uncluttered, and efficient workflow reduces the risk of accidents like spills, burns, or cuts.
 6. **IMPROVES STANDARD OF LIVING:** A well-managed home is more organized, less stressful, and has more resources (time/money) available, which contributes to a higher quality of life.

2: PRINCIPLES OF ENERGY MANAGEMENT (WORK SIMPLIFICATION)

The core of energy management is "work simplification." These are principles designed to make work easier and more efficient.

1. **PLAN YOUR WORK:** Have a clear plan before you start. This includes:
 - **MENU PLANNING:** Know what you will cook for the day or week.

- **SHOPPING LISTS:** Buy all ingredients at once to avoid multiple trips.
 - **ORDER OF WORK:** Decide the logical sequence (e.g., wash all vegetables *before* you start chopping, light the stove *after* all ingredients are ready).
2. **"DOVETAIL" TASKS:** "Dovetailing" means fitting two or more tasks together in the same timeframe.
- **EXAMPLE:** While the rice is boiling (a task that requires waiting), you can wash the salad vegetables (an active task). This uses the "waiting time" productively.
3. **ORGANIZE YOUR WORKSPACE:**
- **"A PLACE FOR EVERYTHING...":** Store all utensils and ingredients in a logical, accessible place.
 - **STORE AT "POINT OF FIRST USE":** Keep pots near the stove, knives and chopping boards near the prep area, spices within easy reach. This eliminates wasted steps.
4. **USE THE RIGHT TOOLS FOR THE JOB:** Use a sharp knife for chopping, a blender for pureeing, a large bowl for mixing. Using the wrong tool (e.g., a small knife for a big yam) wastes time and energy.
5. **USE EFFICIENT WORK METHODS:**
- **SIT TO WORK:** When possible (e.g., peeling vegetables), sit on a comfortable stool to reduce physical strain.
 - **USE BOTH HANDS:** Use both hands for tasks where possible.
 - **USE A TRAY:** Carry multiple items (e.g., salt, spices, oil) to the stove at once on a tray, instead of making multiple trips.
6. **PRACTICE FUEL CONSERVATION:**
- **USE LIDS:** Always cover pots. Water boils faster and stays hot with a lid, saving 20-30% on fuel.
 - **USE A PRESSURE COOKER:** This is the ultimate energy saver for tough foods like beans and meat.
 - **MATCH POT TO BURNER:** Don't use a small pot on a large burner (wastes fuel) or a large pot on a small burner (inefficient).
 - **COOK IN BULK:** It takes less fuel to cook one large pot of stew than to cook five small pots.

3: STEPS IN MANAGING ENERGY

This is the formal process for managing any household task:

1. PLAN:

- **IDENTIFY THE TASK:** (e.g., "Prepare Jollof rice and chicken").
- **SET A GOAL:** (e.g., "Have the meal ready by 1 PM").
- **LIST ALL SUB-TASKS:** (e.g., wash chicken, boil, fry, make stew base, cook rice, etc.).
- **GATHER RESOURCES:** Check that you have all ingredients (rice, chicken, oil, tomatoes) and resources (gas, clean pots).

2. ORGANIZE:

- **SEQUENCE THE TASKS:** Create a logical "Order of Work" (e.g., 1. Wash & boil chicken. 2. While chicken boils, blend peppers. 3. Fry chicken. 4. Use chicken stock/oil to start Jollof...).
- **ASSIGN TASKS (IF IN A GROUP):** (e.g., "Person A chops, Person B cooks").
- **ARRANGE WORKSPACE:** Get out all pots, knives, and bowls needed.

3. IMPLEMENT (CARRY OUT THE PLAN):

- Follow your plan.
- Be flexible. If something goes wrong (e.g., the gas finishes), be ready to adjust.

4. EVALUATE:

- **AFTER THE TASK:** Ask questions.
- "Did I meet my goal?" (Was lunch ready by 1 PM?)
- "Was I successful?" (Did the food taste good?)
- "How tired am I?" (Was I efficient, or am I exhausted?)
- "What would I do differently next time?" (e.g., "Next time, I will blend the peppers the night before to save time").

4: MEANING AND TYPES OF FATIGUE

- **MEANING OF FATIGUE:** Fatigue is a feeling of extreme tiredness, weariness, or exhaustion resulting from physical or mental exertion. It is a signal from the body to rest.

- **TYPES OF FATIGUE:**

1. **PHYSICAL FATIGUE:**

- **CAUSE:** Overuse of muscles. This is the most common type in cooking. Caused by standing for long periods, lifting heavy pots, or repetitive motions (e.g., pounding yam).
- **SOLUTION:** Rest, sitting down, gentle stretching.

2. **MENTAL FATIGUE:**

- **CAUSE:** Overuse of the brain. Caused by stress, concentration, worry, or complex planning (e.g., planning a menu for 50 people, worrying about money).
- **SOLUTION:** Short breaks, changing the task, relaxation, sleep.

3. **PSYCHOLOGICAL FATIGUE (BOREDOM):**

- **CAUSE:** Lack of interest or motivation. Caused by doing the same, monotonous task every single day (e.g., "Ugh, I have to cook stew *again*").
- **SOLUTION:** Varying the routine, playing music, trying new recipes, changing the environment.

EVALUATION

1. Define "Energy Management" in the context of food preparation.
2. List three advantages of practicing good energy management in the kitchen.
3. What is "dovetailing"? Give one clear example you could use when cooking.
4. Explain the four steps in the "management process" (Plan, Organize, Implement, Evaluate).
5. What is "fatigue"? Name and describe the three main types of fatigue.

ASSIGNMENT

1. Write a detailed "Plan of Work" (Order of Work) for preparing a simple meal of Boiled Yam and Egg Stew. The plan should show at least *two* examples of dovetailing tasks.
2. "Work Simplification" is key to reducing fatigue. Write a 400-word essay on five principles of work simplification.

3. Draw a diagram of your kitchen at home (or an ideal kitchen). Show the "work triangle" (Stove, Sink, Refrigerator). Using arrows, show three examples of *wasted energy* (bad workflow) and explain how you would fix it.
4. A pressure cooker is an excellent energy-saving device. Research and explain the scientific principle of *how* a pressure cooker works to cook food faster.
5. Observe someone cooking a meal (or observe yourself). Identify three instances that lead to fatigue (e.g., poor posture, disorganized space, psychological boredom). Write a report on your findings.



WEEK 2: KITCHEN EQUIPMENT AND UTENSILS

TOPIC: KITCHEN EQUIPMENT AND UTENSILS

INTRODUCTION: A well-equipped kitchen is essential for efficient, safe, and enjoyable food preparation. Kitchen equipment and utensils are the "tools of the trade." Understanding their classification, selection, and care is a fundamental skill for any student of food and nutrition.



1: MEANING AND CLASSIFICATION

- **MEANING:**
 - **KITCHEN EQUIPMENT:** Refers to the larger, more significant items in a kitchen. They are often mechanical or electrical. (e.g., cooker, refrigerator, sink).

- **KITCHEN UTENSILS:** Refers to the smaller, handheld tools used in food preparation. (e.g., knives, spoons, bowls, pots).
- **CLASSIFICATION:**
 1. **LARGE EQUIPMENT (FIXTURES):**
 - **MEANING:** Large, expensive items that are often "fixed" in place and are considered part of the kitchen's structure.
 - **EXAMPLES:** Cooker (gas or electric), Oven, Refrigerator/Freezer, Kitchen Sink, Storage Cabinets, Countertops.
 2. **SMALL EQUIPMENT (APPLIANCES):**
 - **MEANING:** Smaller electrical or mechanical items that are usually portable.
 - **EXAMPLES:** Blender, Food processor, Electric mixer, Toaster, Microwave oven, Pressure cooker, Electric kettle.
 3. **KITCHEN UTENSILS (TOOLS):**
 - **MEANING:** Small, non-mechanical tools used for specific tasks.
 - **CLASSIFIED BY USE:**
 - **COOKING:** Pots (saucepans, stewpots), Frying pans, Pot lids.
 - **PREPARATION:** Knives (chef's, paring), Chopping board, Mixing bowls, Sieves, Graters, Measuring cups/spoons, Whisk, Spatula, Wooden spoon.
 - **SERVING:** Serving spoons, Ladle, Trays, Dishes, Plates, Cutlery.
 - **CLASSIFIED BY MATERIAL:**
 - **STAINLESS STEEL:** (Pots, pans, knives, bowls) - Durable, hygienic, easy to clean.
 - **ALUMINIUM:** (Pots, pans) - Lightweight, cheap, good heat conductor, but can react with acidic foods.
 - **WOOD:** (Spoons, chopping boards) - Does not scratch pans, but can absorb germs if not cleaned properly.
 - **PLASTIC/NYLON:** (Spatulas, spoons) - Safe for non-stick pans, but can melt.

- **GLASS/CERAMIC:** (Bowls, baking dishes) - Hygienic, but breaks easily.

2: FACTORS TO CONSIDER IN SELECTION

Choosing the right equipment is a long-term investment.

1. NEEDS OF THE FAMILY:

- A large family needs a large refrigerator and large pots.
- A small family or single person needs smaller items.

2. COST (BUDGET):

- Buy the best quality you can *afford*.
- Consider *running costs* (an energy-efficient fridge may cost more upfront but save money on electricity bills).

3. DURABILITY:

- Is it well-made? Are the handles secure?
- Stainless steel is more durable than plastic.

4. EASE OF CLEANING (HYGIENE):

- Avoid items with "dirt traps" (crevices, complex parts).
- Stainless steel and glass are very hygienic.
- Is it "dishwasher safe"?

5. EASE OF USE:

- Is it simple to operate?
- Is it too heavy (e.g., a heavy pot)?
- Are the controls clear?

6. VERSATILITY (MULTI-PURPOSE):

- Can one item do many jobs? (e.g., a food processor can chop, blend, and mix).
- A good chef's knife can do 90% of all cutting jobs.

7. SPACE:

- Do you have room for it? This is crucial for large equipment.

8. SAFETY:

- Does it have safety features (e.g., a secure lid on a blender)?

- Is it made from "food-grade" materials?

3: USE, CARE, AND STORAGE OF EQUIPMENT

Proper care extends the life of your equipment and ensures food safety.

- **GENERAL RULES:**

1. **READ THE MANUAL:** Always read the manufacturer's instructions for new appliances.
2. **CLEAN AFTER USE:** Clean all equipment promptly after use. Do not leave dirty dishes overnight.
3. **USE FOR INTENDED PURPOSE:** Don't use a knife as a can opener or a glass bowl to cook on the stove.
4. **STORE CORRECTLY:** Store in a clean, dry place. Knives should be stored in a block or on a rack, not jumbled in a drawer (this dulls them and is dangerous).

- **SPECIFIC CARE BY MATERIAL:**

- **STAINLESS STEEL:** Wash with warm, soapy water. Very durable.
- **ALUMINIUM:** Wash with soap and water. *Do not* use harsh, abrasive cleaners (like steel wool), which can scratch it. Acidic foods (like tomatoes) can "pit" the surface; wash immediately after.
- **WOOD:** (Spoons, boards) - Wash quickly and dry *immediately*. **Never** soak wooden items in water, as this will cause them to swell, crack, and harbour germs.
- **NON-STICK PANS:** (Teflon) - **Never** use metal utensils (spoons, forks) on them; use only wood or plastic. Wash gently with a soft sponge.
- **ELECTRICAL APPLIANCES:** (Blender, mixer) - **Never** submerge the motor base in water. Wipe the base with a damp cloth. Wash the detachable parts (jug, blades) separately.
- **KNIVES:** Wash and dry by hand. Do not leave them soaking in the sink. Store them in a block or magnetic strip.

EVALUATION

1. What is the main difference between "kitchen equipment" and "kitchen utensils"?
2. Classify the following items: (a) Refrigerator, (b) Whisk, (c) Blender.
3. List five key factors to consider when selecting kitchen equipment.
4. You are buying a new pot. Explain the pros and cons of an "Aluminium" pot vs. a "Stainless Steel" pot.
5. What is the *most important* rule for the care of (a) wooden spoons and (b) non-stick frying pans?

ASSIGNMENT

1. Conduct an "inventory" of your kitchen at home. Create a table with three columns: (1) Large Equipment, (2) Small Appliances, (3) Utensils. List at least five items for each.
2. Write a 400-word essay on "Factors to consider when buying a new Refrigerator/Freezer."
3. A good "Chef's Knife" is a cook's best friend. Research and explain the difference between a "Chef's Knife," a "Paring Knife," and a "Serrated Knife."
4. Design a poster for a school Home Economics class titled "How to Care for Wooden Utensils." It must include at least 3 "DOs" and 3 "DON'Ts."
5. Research "stainless steel." What is it an alloy of? Why is it "stainless" (i.e., why doesn't it rust)?

WEEK 3: LABOUR SAVING DEVICES

TOPIC: LABOUR SAVING DEVICES

INTRODUCTION: Labour saving devices are a specific category of kitchen equipment designed to reduce human effort (labour) and save time. They replace slow, difficult, or repetitive manual tasks with mechanical or electrical power, directly contributing to energy management and work simplification.



1: MEANING AND USES OF LABOUR SAVING DEVICES

- **MEANING:** A labour saving device is any tool, appliance, or machine that is designed to make a task easier and faster, reducing the amount of physical work (labour) required.
- **USES (EXAMPLES):**
 - **BLENDER / LIQUIDIZER:**
 - **MANUAL TASK IT REPLACES:** Grinding pepper/beans on a grinding stone; mashing.
 - **USE:** Creates smooth purées, blends liquids (for smoothies), and grinds wet ingredients (e.g., peppers for stew).
 - **FOOD PROCESSOR:**
 - **MANUAL TASK IT REPLACES:** Slicing, chopping, grating, mixing, kneading.
 - **USE:** A highly versatile tool. Can chop onions in 2 seconds, grate a block of cheese, slice cucumbers, or mix pastry dough.
 - **ELECTRIC MIXER (HAND OR STAND):**
 - **MANUAL TASK IT REPLACES:** Beating eggs with a fork; creaming butter and sugar with a wooden spoon; kneading dough.
 - **USE:** Incorporates air for baking (creaming, whisking) or develops gluten (kneading with a dough hook).
 - **PRESSURE COOKER:**
 - **MANUAL TASK IT REPLACES:** Boiling tough food (beans, meat) for many hours.
 - **USE:** Cooks food in a fraction of the time, saving human time *and* fuel.
 - **MICROWAVE OVEN:**
 - **MANUAL TASK IT REPLACES:** Reheating food on a stove; defrosting frozen food.
 - **USE:** Extremely fast reheating and defrosting.
 - **RICE COOKER:**
 - **MANUAL TASK IT REPLACES:** Watching a pot of rice to ensure it doesn't burn.
 - **USE:** Cooks rice perfectly and automatically switches to "keep warm" mode, requiring no supervision.

- **YAM POUNDER (MACHINE):**

- **MANUAL TASK IT REPLACES:** The extremely strenuous act of pounding yam in a mortar.
- **USE:** Produces pounded yam with minimal effort.

2: FACTORS TO CONSIDER IN SELECTION

This is similar to Week 2, but with a special focus on the *function* of the device.

- 1. FUNCTION (WHAT DOES IT DO?):**

- Is it a *multi-tasker* (like a food processor) or a *single-tasker* (like a toaster)? A multi-tasker often offers better value.

- 2. EASE OF ASSEMBLY AND WASHING:**

- This is critical. A device that is difficult to assemble or has 20 small parts to wash will *not* save you labour. You will avoid using it. Look for simple, dishwasher-safe parts.

- 3. COST AND BUDGET:** These devices can be expensive.

- 4. DURABILITY (MOTOR QUALITY):** The "motor" is the heart of the appliance. A cheap blender with a weak motor will burn out quickly.

- 5. SPACE (STORAGE):** Where will you keep it? A large stand mixer or food processor needs significant counter or cupboard space.

- 6. NOISE:** Some devices (blenders, grinders) can be very loud.

- 7. SAFETY FEATURES:** A good food processor or blender will *not* start unless the lid is locked in place.

- 8. AVAILABILITY OF POWER:** These devices are useless if you do not have a stable electricity supply.

3: CARE AND MAINTENANCE

- **CARE:**

- 1. NEVER IMMERSE THE MOTOR BASE IN WATER:** This is the golden rule. It will cause an electric shock and destroy the appliance. Wipe the base with a damp cloth.

2. **WASH PARTS PROMPTLY:** Do not let food (especially tomato) dry on the parts; it becomes difficult to clean.
 3. **HANDLE BLADES WITH CARE:** They are extremely sharp. Clean them with a brush, not your fingers.
 4. **DO NOT OVERFILL:** Do not fill a blender jug past the "MAX" line. This puts strain on the motor and can cause leaks.
 5. **DO NOT OVERHEAT:** Do not run a blender or mixer for 10 minutes straight. Run it in short bursts to let the motor rest and prevent burnout.
 6. **USE THE CORRECT ATTACHMENT:** Use a "dough hook" for bread, not a "whisk."
- **MAINTENANCE:**
 1. **STORE PROPERLY:** Store in a clean, dry place, away from steam or water.
 2. **CHECK CORDS:** Regularly check the electrical cord for any cuts or damage.
 3. **SHARPEN BLADES:** Some blender blades can be sharpened, or a replacement can be bought.
 4. **SERVICE (PROFESSIONAL):** If a device is making a strange noise or sparking, unplug it immediately and take it to a qualified repair person.

EVALUATION

1. What is a "labour saving device"?
2. What *manual task* does a (a) blender and (b) electric mixer replace?
3. What is the *most important* factor to consider regarding "ease of cleaning" when buying a food processor?
4. What is the "golden rule" for cleaning any electrical appliance with a motor base?
5. Why is it a bad idea to run a blender continuously for a long time without a break?

ASSIGNMENT

1. Choose *one* labour saving device (e.g., pressure cooker, microwave, blender). Write a 400-word report on: (a) Its history/invention, (b) The scientific principle of *how* it works, and (c) Its impact on home cooking.

2. A food processor and a blender seem similar, but they do different jobs. Research and write a detailed comparison, explaining what a food processor is *good at* (e.g., chopping) and what a blender is *good at* (e.g., puréeing).
3. Go to an online store (like Jumia/Konga) or a local appliance shop. Find a "cheap" blender and an "expensive" blender. Compare their features (e.g., motor power in watts, jug material, number of blades). Write one paragraph on why the expensive one might be a better long-term investment.
4. Design a "Safety & Care" poster for a blender. It must include 3 "DOs" and 3 "DON'Ts."
5. What is a "Stand Mixer"? How is it different from a "Hand Mixer"? Which one would be more useful for a professional baker, and why?



WEEK 4: SPECIAL NUTRITIONAL NEEDS I

TOPIC: NUTRITIONAL NEEDS OF DIFFERENT LIFECYCLE GROUPS

INTRODUCTION: Nutritional needs are not static; they change dramatically throughout the human lifecycle. A "balanced diet" must be adapted to meet the specific physiological demands of each stage, from the rapid growth of infancy to the maintenance and repair of old age.



1: INFANTS (0-1 YEAR)

- **PHYSIOLOGICAL NEEDS:**
 - **EXTREMELY RAPID GROWTH:** A baby's weight *triples* in the first year. This requires a very high-energy and high-protein diet *relative to their size*.

- **HIGH ENERGY NEEDS:** The highest BMR (Basal Metabolic Rate) of any life stage.
- **IMMATURE DIGESTIVE SYSTEM:** Cannot handle complex foods, fibre, or strong flavours.
- **IMMATURE KIDNEYS:** Cannot handle high loads of salt or protein.
- **BRAIN DEVELOPMENT:** Requires high amounts of fat and specific fatty acids (DHA).
- **MEAL PLANNING (NUTRITIONAL REQUIREMENTS):**
 - **0-6 MONTHS: EXCLUSIVE BREASTFEEDING.**
 - Breast milk is the "perfect food," providing all energy, protein, fat (DHA), vitamins, water, and *antibodies* (passive immunity).
 - Infant formula is the only safe alternative. Cow's milk is **unsuitable** (too much protein/salt, not enough iron/fat).
 - **6+ MONTHS: WEANING.**
 - **MEANING:** The gradual introduction of "complementary" solid foods *alongside* breast milk or formula.
 - **WHY AT 6 MONTHS?** The baby's iron stores (from birth) run out and must be replenished from food.
 - **FIRST FOODS:** Should be smooth, single-ingredient purées. (e.g., iron-fortified baby cereal, mashed yam, puréed fruit/vegetables).
 - **PRINCIPLES:** Introduce new foods one at a time. No salt, no sugar.

2: CHILDREN (1-12 YEARS)

- **PHYSIOLOGICAL NEEDS:**
 - **SLOWER GROWTH:** Growth is slower and occurs in "spurts" (not as fast as infancy).
 - **HIGH ACTIVITY:** Very active, so energy needs are high.
 - **SMALL STOMACHS:** Cannot eat large meals.
 - **BONE/TEETH DEVELOPMENT:** High need for **Calcium** and **Vitamin D**.
- **MEAL PLANNING:**

- **"NUTRIENT-DENSE" FOODS:** Since they eat small portions, every bite must count.
- **AVOID "EMPTY CALORIES":** Limit sugary drinks, sweets, and processed snacks.
- **PLAN 5-6 SMALL MEALS:** Three main meals and 2-3 healthy snacks (e.g., fruit, yogurt) are better than 3 large meals.
- **MAKE FOOD APPEALING:** Use colours, cut food into fun shapes.
- **"FOOD FADDISM" / NEOPHOBIA:** Children often go through "fussy" phases. It is the parent's job to *offer* healthy food; it is the child's job to *decide* how much to eat.

3: ADOLESCENTS (13-19 YEARS)

- **PHYSIOLOGICAL NEEDS:**

- **RAPID GROWTH SPURT:** The second-fastest growth period (after infancy).
- **HIGH ENERGY NEEDS:** Energy needs are at their peak, often higher than adults.
- **HIGH PROTEIN NEEDS:** For muscle and tissue growth.
- **KEY MINERALS:**
 - **CALCIUM & VITAMIN D:** For "Peak Bone Mass." The bones built now must last a lifetime.
 - **IRON:** Needs increase dramatically, especially for girls (due to menstruation). Iron-deficiency anaemia is very common.
- **PEER PRESSURE:** Food choices are heavily influenced by friends, social media, and convenience (e.g., fast food).

- **MEAL PLANNING:**

- **FOCUS ON SNACKS:** Adolescents get a large portion of their calories from snacks. Ensure snacks are healthy (e.g., boiled egg, nuts, fruit, yogurt) instead of "empty."
- **IRON-RICH FOODS:** Encourage intake of meat, poultry, fish, and dark leafy greens (with a source of Vitamin C to aid absorption).
- **CALCIUM-RICH FOODS:** Milk, yogurt, cheese, fortified cereals.

- **LIMITING:** Sugary drinks, energy drinks, and fast food.

4: ADULTS (20-64 YEARS)

- **PHYSIOLOGICAL NEEDS:**

- **MAINTENANCE & REPAIR:** Growth has stopped. Nutrients are needed for maintaining the body, repairing tissues, and fuelling work/activity.
- **ENERGY NEEDS DECLINE:** BMR slowly decreases with age. Energy needs are highly *variable* and depend on **activity level**. (A construction worker needs far more than an office worker).
- **RISK OF CHRONIC DISEASE:** This is the stage where poor diet can lead to obesity, Type 2 Diabetes, and heart disease.

- **MEAL PLANNING:**

- **MAINTENANCE (ENERGY BALANCE):** Calories "in" must equal calories "out" to prevent weight gain.
- **FOCUS ON PREVENTION:**
 - **FIBRE:** High-fibre foods (vegetables, whole grains) for gut health.
 - **LIMIT:** Saturated fat, sugar, and sodium (salt).
- **ACTIVITY:** Diet must be balanced with regular physical activity.

5: PREGNANT AND LACTATING MOTHERS

- **PREGNANT MOTHERS:**

- **PHYSIOLOGICAL NEEDS:** "Eating for two" is a myth. It's "eating for one and a half."
- **ENERGY:** Only a small increase in calories is needed (approx. 300 extra kcal/day, mostly in the 2nd/3rd trimesters).
- **KEY NUTRIENTS:**
 - **FOLIC ACID (FOLATE):** Critical *before* and in the *first few weeks* of pregnancy to prevent "neural tube defects" (e.g., Spina Bifida).
 - **IRON:** Blood volume expands, so needs increase dramatically.
 - **PROTEIN & CALCIUM:** For building the baby's tissues and bones.

- **FOODS TO AVOID:** Alcohol, high-mercury fish, raw/undercooked meat, unpasteurized cheese (risk of Listeria).
- **LACTATING MOTHERS (BREASTFEEDING):**
 - **PHYSIOLOGICAL NEEDS:**
 - **HIGH ENERGY NEEDS:** Producing milk is very energy-intensive, requiring approx. 500 extra kcal/day (more than pregnancy).
 - **HIGH FLUID NEEDS:** Must drink plenty of water to produce milk.
 - **PROTEIN & CALCIUM:** Needs remain high to ensure milk quality.
- **MEAL PLANNING:**
 - Nutrient-dense foods. Lots of water. Continue taking prenatal vitamins.

6: THE AGED / ELDERLY (65+ YEARS)

- **PHYSIOLOGICAL NEEDS:**
 - **LOWER ENERGY NEEDS:** BMR is lower, and activity is often reduced. Risk of weight gain is high.
 - **HIGH NUTRIENT NEEDS:** Needs for *nutrients* (protein, calcium, vitamins) do *not* decrease. This means they must eat *nutrient-dense* foods.
 - **CHALLENGES:**
 - **DIGESTION:** Poorer digestion, constipation (need fibre).
 - **TEETH/DENTURES:** Difficulty chewing (may need softer foods).
 - **APPETITE/TASTE:** Reduced sense of taste and smell can lead to poor appetite.
 - **OSTEOPOROSIS:** High risk of brittle bones; **Calcium** and **Vitamin D** are critical.
- **MEAL PLANNING:**
 - **SOFT, MOIST FOODS:** (e.g., stews, soups, boiled yam, fufu).
 - **HIGH-FIBRE:** (e.g., oats, vegetables, beans).
 - **FLAVOURFUL:** Use herbs and spices (not salt) to boost flavour and appetite.
 - **STAY HYDRATED:** The sense of thirst also declines; they must be encouraged to drink water.

EVALUATION

1. Why is "Exclusive Breastfeeding" recommended for infants aged 0-6 months?
2. What is "weaning," and what is the key nutrient (which runs out at 6 months) that weaning foods must provide?
3. Name two key minerals that are critical for adolescents and explain *why* each is needed.
4. "Eating for two" is a myth. What *is* the most important nutrient for a woman to take *before* and in the *early weeks* of pregnancy, and why?
5. List three physiological *challenges* that can affect the nutritional status of an elderly person.

ASSIGNMENT

1. Write a 400-word essay comparing the nutritional needs of a 2-year-old child with those of an 18-year-old adolescent.
2. What is "Iron-Deficiency Anaemia"? Why are adolescent girls and pregnant women at the highest risk? List 5 iron-rich foods.
3. Design a one-day "meal plan" (breakfast, lunch, dinner, 2 snacks) for a *pregnant woman* in her second trimester. Ensure it is rich in folate, iron, and protein.
4. Design a one-day "meal plan" for an *elderly person* who has difficulty chewing. All foods must be nutrient-dense and soft.
5. What is "Peak Bone Mass"? Research this concept and explain why the adolescent years are the most critical "window" for lifelong bone health.

WEEK 5: MIDTERM TEST

TOPIC: ASSESSMENT OF WEEKS 1–4

INTRODUCTION: This week is dedicated to a formal assessment of all concepts covered in the first part of the term. The test will evaluate your understanding of energy management, kitchen equipment, and the nutritional needs of lifecycle groups.

SCOPE:

- **WEEK 1: ENERGY MANAGEMENT:** Meaning, advantages, principles (work simplification, dovetailing), steps in management (P-O-I-E), fatigue (types).
- **WEEK 2: KITCHEN EQUIPMENT:** Classification (large, small, utensils), selection factors (cost, durability, hygiene), care by material (wood, non-stick, aluminium).
- **WEEK 3: LABOUR SAVING DEVICES:** Meaning, examples (blender, pressure cooker), selection factors (cleaning, motor), care (motor base, overheating).
- **WEEK 4: SPECIAL NUTRITIONAL NEEDS I:** Detailed physiological needs and meal planning principles for Infants (weaning), Children, Adolescents (iron/calcium), Adults, Pregnant women (folate/iron), and the Aged.

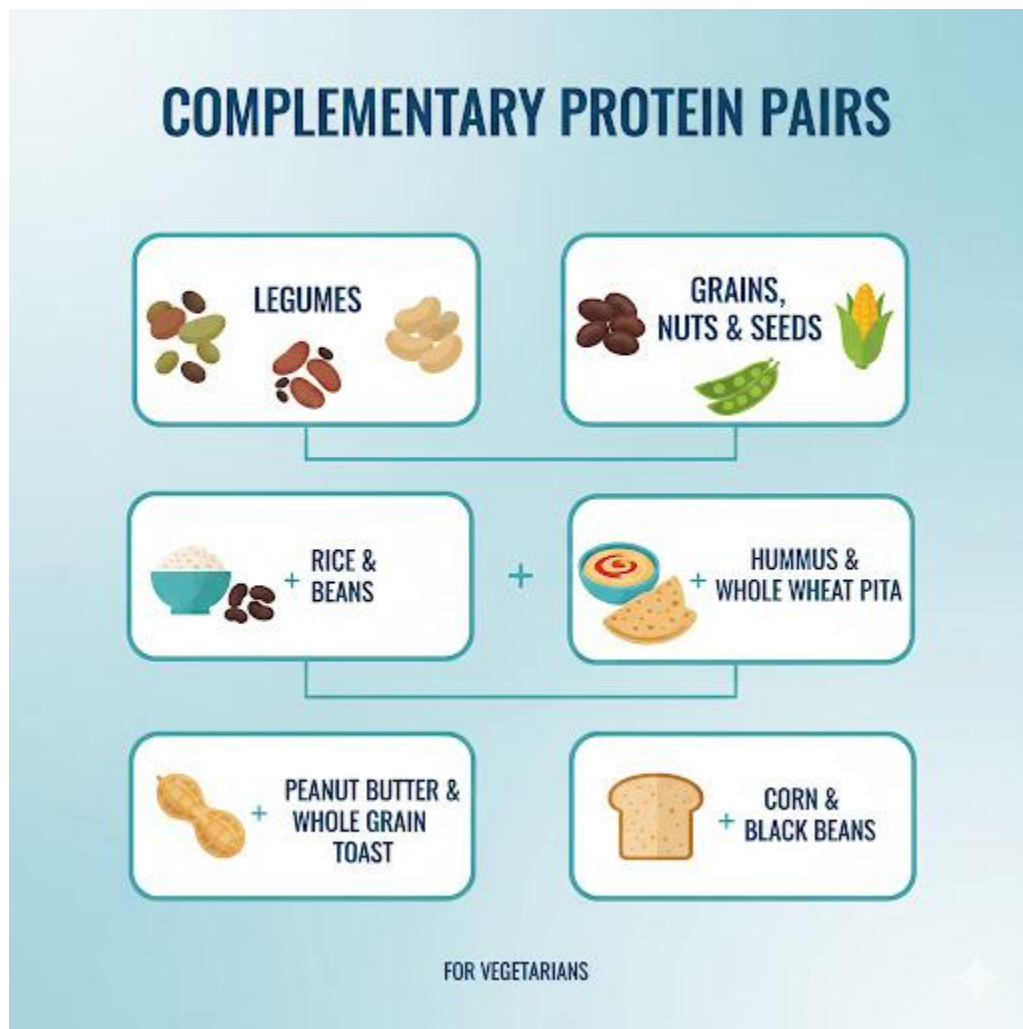
FORMAT:

- **SECTION A: MULTIPLE CHOICE QUESTIONS (20 MARKS)** - Testing definitions and key facts.
- **SECTION B: SHORT ANSWER QUESTIONS (30 MARKS)** - Testing explanations (e.g., "Explain 3 principles of work simplification," "List 3 challenges for an elderly person").
- **SECTION C: ESSAY/LONG ANSWER QUESTIONS (50 MARKS)** - (e.g., "Discuss the physiological needs and meal planning principles for an adolescent," "You are buying a new blender; discuss 5 factors you must consider").

WEEK 6: SPECIAL NUTRITIONAL NEEDS II

TOPIC: CONTINUATION OF MEAL PLANNING (SPECIAL GROUPS)

INTRODUCTION: Last week, we covered the nutritional needs of the *lifecycle*. This week, we will continue by exploring other groups who have special dietary needs based on their *health status* or *lifestyle choices*.



1: INVALIDS (CONVALESCENTS)

- **MEANING:**

- **INVALID:** A person who is sick.
- **CONVALESCENT:** A person who is *recovering* from an illness or surgery.
- **PHYSIOLOGICAL NEEDS:**
 - **INCREASED NUTRITIONAL NEEDS:** The body is fighting infection and/or repairing tissue, which requires *more* protein, vitamins (especially C and Zinc for healing), and energy.
 - **POOR APPETITE:** Illness often causes a loss of appetite, nausea, or weakness.
 - **DIGESTION:** The digestive system may be weak or sensitive.
- **MEAL PLANNING (THE GOALS):**
 1. **HIGHLY NUTRITIOUS:** Must be nutrient-dense.
 2. **EASILY DIGESTIBLE:** Avoid heavy, oily, or spicy foods.
 3. **SERVED IN SMALL, FREQUENT MEALS:** A large plate is overwhelming. 6 small snacks/meals are better.
 4. **SERVED ATTRACTIVELY:** Use nice plates and add colour (e.g., a sprig of parsley) to tempt the poor appetite.
 5. **FLUIDS:** Crucial for preventing dehydration (especially with fever).
- **SUITABLE DISHES:**
 - **LIQUIDS:** Fruit juice, light soups (pepper soup), broths, pap (ogi).
 - **LIGHT/SOFT:** Boiled yam, poached eggs, steamed fish, yogurt, ripe fruits.
- **FOODS TO AVOID:** Fried foods, oily stews, processed meats, strong-smelling foods.

2: VEGETARIANS

- **MEANING:** People who do not eat meat or fish, for health, religious, ethical, or environmental reasons.
- **TYPES:**
 1. **LACTO-OVO VEGETARIAN:** Eats dairy (lacto) and eggs (ovo). *Does not* eat meat, poultry, or fish. This is the most common type.
 2. **LACTO-VEGETARIAN:** Eats dairy. *Does not* eat eggs, meat, poultry, or fish.

- 3. **VEGAN:** Eats *no animal products* of any kind. This includes honey, dairy, and eggs. This is the strictest form.
- **NUTRITIONAL NEEDS (AREAS FOR CAUTION):**
 - **PROTEIN:** It is easy to get protein. However, plant proteins (except soy) are "incomplete."
 - **SOLUTION: COMPLEMENTARY PROTEINS.** This means eating two different plant proteins together to get all essential amino acids.
 - **EXAMPLES:** Rice and Beans, Yam and Egusi Soup (without fish), Bread and Peanut Butter.
 - **IRON:** Plant-based (non-haem) iron is poorly absorbed.
 - **SOLUTION:** Eat with Vitamin C (e.g., beans with a tomato stew) and avoid drinking tea with meals.
 - **VITAMIN B12:** This is the *biggest* risk, especially for vegans, as B12 is not found in plants.
 - **SOLUTION:** Must consume fortified foods (e.g., fortified cereal, soy milk) or take a B12 supplement.

3: ATHLETES (HIGH PHYSICAL ACTIVITY)

- **PHYSIOLOGICAL NEEDS:**
 - **HIGH ENERGY:** Extremely high calorie needs to fuel performance.
 - **CARBOHYDRATES (THE PRIMARY FUEL):** This is the most important nutrient. Carbs are stored in muscles as *glycogen* and are the main fuel for high-intensity exercise.
 - **PROTEIN:** Increased protein needs for muscle repair (not as fuel).
 - **FLUIDS (HYDRATION):** This is critical. Dehydration severely impacts performance.
- **MEAL PLANNING:**
 - **BEFORE EXERCISE:** A light, high-carb, low-fat snack (e.g., a banana, bread).
 - **DURING EXERCISE (IF LONG):** Simple sugars for quick energy (e.g., sports drinks, gels).

- **AFTER EXERCISE (RECOVERY):** A meal containing both **Carbs** (to refill glycogen) and **Protein** (to repair muscle). (e.g., Jollof rice and chicken, a milk-based drink).
- **HYDRATION:** Drink water before, during, and after.

EVALUATION

1. What is a "convalescent," and what are the three main goals of meal planning for them?
2. What is the difference between a "Lacto-Ovo Vegetarian" and a "Vegan"?
3. What is the *most critical* vitamin that a Vegan must get from fortified foods or supplements?
4. Explain the concept of "Complementary Proteins" and give one example.
5. What is the most important *fuel* nutrient for an athlete, and what is the most important *recovery* nutrient combination?

ASSIGNMENT

1. Create a one-day "soft food" meal plan (6 small meals) for an invalid recovering from a fever.
2. Write a 400-word essay on the "Pros and Cons of a Vegetarian Diet."
3. A vegan student is your friend. Create a list of 5 Nigerian-friendly meals/snacks they can eat (e.g., "Akara and Pap," "Roast corn and pear").
4. What is "Carbohydrate Loading"? Research this practice (used by marathon runners) and explain it in one paragraph.
5. What is "Osteoporosis"? Research this disease (common in the elderly, as seen in Week 4) and explain how diet can help prevent it.

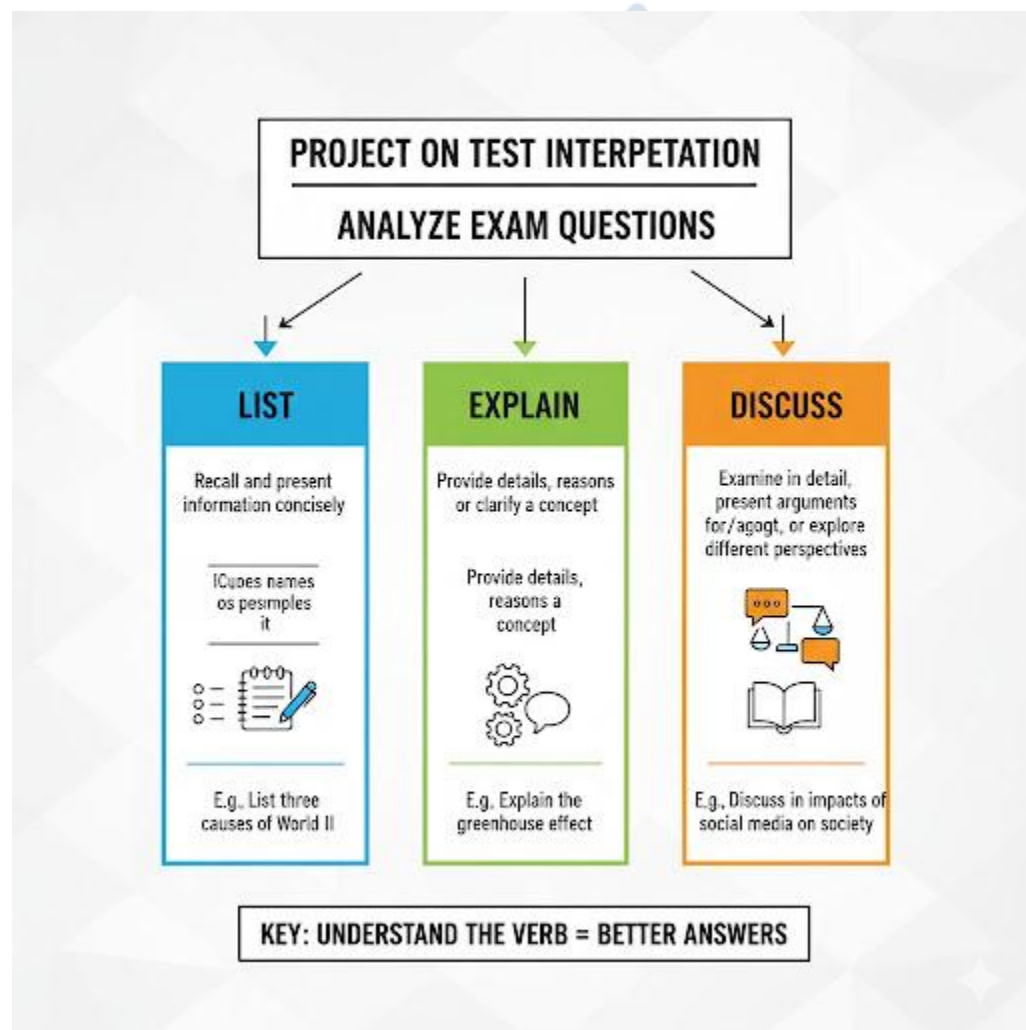
WEEK 7: MIDTERM BREAK



WEEK 8: PROJECT ON TEST INTERPRETATION

TOPIC: PRACTICE IN INTERPRETING EXAM QUESTIONS

INTRODUCTION: In Food and Nutrition (and all subjects), getting good grades is not just about *knowing* the content; it is about *understanding* what the examiner is asking you to do. Many students fail exams not because they don't know the answer, but because they don't answer the *specific question asked*. This week is a practical project in "learning how to be tested."



1: UNDERSTANDING INSTRUCTIONAL VERBS

The "instructional verb" (or "action word") is the *most important* word in the question. It tells you the *level of detail* required.

- **THE "LOW-LEVEL" VERBS (MEMORY):**

- **LIST / NAME / STATE:** Give a simple list. No explanation needed.
 - *Example:* "List three types of poultry." (Answer: "Chicken, Turkey, Duck").
- **DEFINE:** Give the formal, "dictionary" meaning of a term.
 - *Example:* "Define 'fatigue'."
- **OUTLINE:** Give the main points in a logical order, without detail.

- **THE "MID-LEVEL" VERBS (UNDERSTANDING):**

- **DESCRIBE:** Give a detailed account of something, step-by-step or with its characteristics.
 - *Example:* "Describe the 'break test' for egg freshness."
- **EXPLAIN:** Give a detailed account, but this time you *must* include the "Why" or "How."
 - *Example:* "Explain *why* an old hen is unsuitable for roasting." (Answer: "It is tough... needs moist heat... to break down collagen...").
- **ILLUSTRATE:** Explain, but use *examples* or *diagrams* to make your point clear.
 - *Example:* "Illustrate the culinary uses of eggs." (You must give examples like "Binding, e.g., in meatballs...").

- **THE "HIGH-LEVEL" VERBS (ANALYSIS & EVALUATION):**

- **COMPARE:** Show the *similarities* between two things.
- **CONTRAST / DIFFERENTIATE:** Show the *differences* between two things.
- **COMPARE AND CONTRAST:** Show *both* similarities and differences.
 - *Example:* "Compare and contrast pasteurization and UHT."
- **DISCUSS:** This is a major essay question. You must explore the topic from multiple angles (e.g., pros and cons, causes and effects).
 - *Example:* "Discuss the nutritional needs of adolescents."
- **EVALUATE / ASSESS:** Make a *judgement*. Give your opinion based on evidence.

- *Example:* "Evaluate the suitability of a vegan diet for an infant." (Answer: "It is unsuitable... high risk of B12 deficiency... etc.").

2: PROJECT - DECONSTRUCTING EXAM QUESTIONS

This is the main project for the week.

• PART 1: ANALYSIS (PRACTICE):

- *TASK:* Analyse the following three questions.
- *Question 1:* "State three factors in choosing kitchen equipment."
- *Question 2:* "Explain three factors in choosing kitchen equipment."
- *Question 3:* "Discuss the factors to consider when purchasing a new refrigerator."
- *YOUR ANALYSIS:*
 - **Q1:** Is a "List" question. It needs 3 simple points (e.g., "Cost, Durability, Space"). (Worth 3-6 marks).
 - **Q2:** Is an "Explain" question. It needs 3 points *with* explanations. (e.g., "Cost, because you must stay within your budget. Durability, because a well-made item..."). (Worth 9-12 marks).
 - **Q3:** Is a "Discuss" question. It needs a full essay, with an introduction, multiple points (cost, space, energy, durability, family size), and a conclusion. (Worth 20-25 marks).

• PART 2: WRITING RESPONSES (PROJECT):

- *TASK:* You will be given a set of past exam questions. Your project is to create a "perfect answer" key.
- **STEP 1:** For each question, *first* identify the instructional verb (e.g., "Define") and the key topic (e.g., "Weaning").
- **STEP 2:** Write a complete, accurate, and full-mark answer that *perfectly matches* the instruction. (i.e., don't "explain" if it only asks you to "list").

EVALUATION

1. What is an "instructional verb" in an exam question?
2. What is the key difference between a "List" question and an "Explain" question?

3. What is the key difference between "Compare" and "Contrast"?
4. If a question asks you to "Discuss," what kind of answer is required?
5. Analyse this question: "Evaluate the challenges in planning meals for the elderly." What is the instructional verb, and what is the key topic?

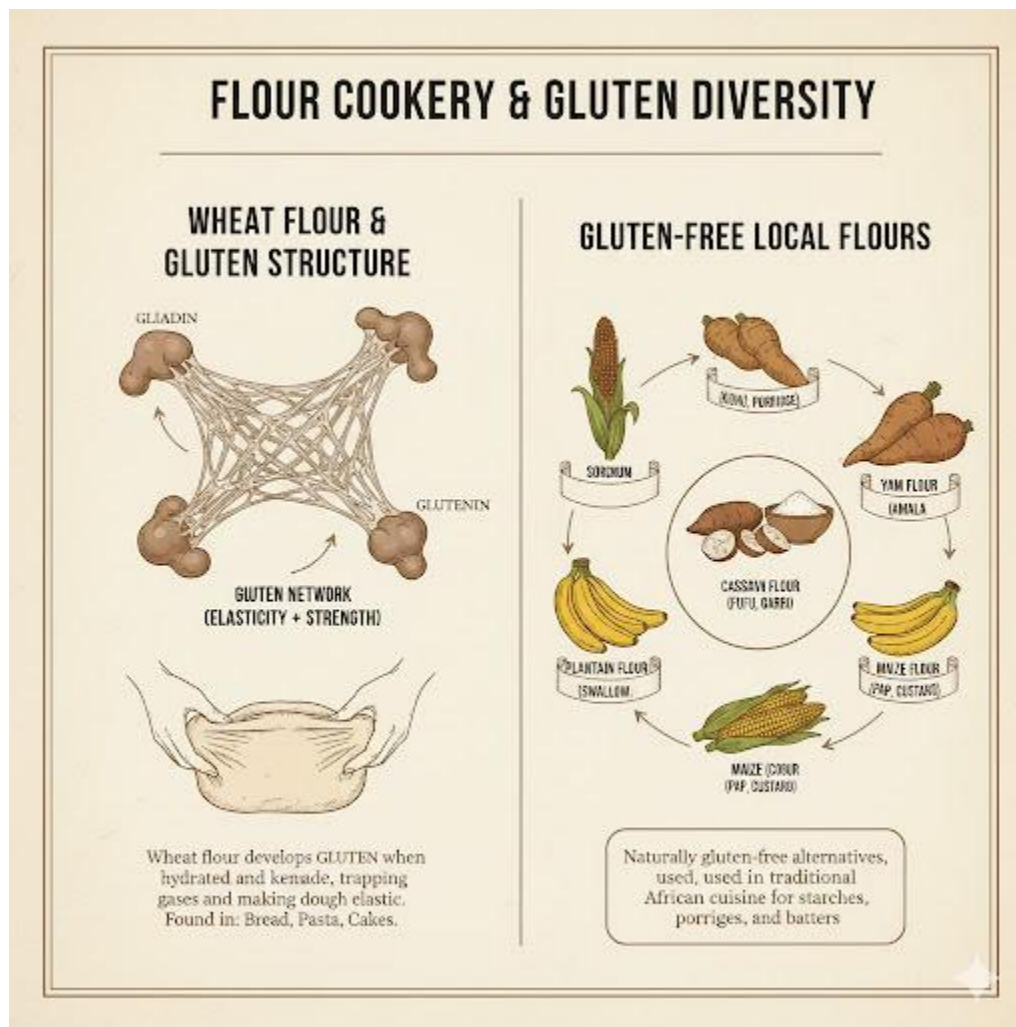
ASSIGNMENT (THE PROJECT)

1. **PART A:** Find (or your teacher will provide) 5-10 past exam questions from your SS1 or SS2 notes.
2. Create a "Test Interpretation" document. For each question:
 - Write the full question.
 - Identify and **bold** the instructional verb.
 - Identify and *underline* the key topic(s).
 - Write one sentence explaining *what* is required (e.g., "This requires a detailed essay on the pros and cons...").
3. **PART B:** Choose *three* of these questions (one "List," one "Explain," and one "Discuss") and write the full, perfect answer for them.
4. **PART C:** Write a 300-word "reflection" on the most common *mistake* you think students make when answering exam questions.
5. What does the acronym "P-O-I-E" (from Week 1) stand for? (This is a revision question).

WEEK 9: FLOUR COOKERY

TOPIC: FLOUR COOKERY

INTRODUCTION: Flour is a fine powder ground from cereal grains, starchy roots, or other plants. It is the fundamental ingredient in baking and a staple product used globally for making bread, pastries, cakes, and as a thickener.



1: MEANING AND TYPES OF FLOUR

- **MEANING:** Flour is the powder produced by milling (grinding) raw grains or roots.

- **WHEAT FLOUR:**

- The most common type, due to a protein called **GLUTEN**.
- **Gluten:** This is an "elastic" protein that forms when wheat flour is mixed with water. It traps air (from yeast or baking powder) and gives bread its chewy, elastic structure.
- **TYPES OF WHEAT FLOUR:**
 - **Bread Flour (Hard Wheat):** High in protein (gluten). Used for yeast breads.
 - **All-Purpose Flour (Plain Flour):** Medium protein. Used for everything (cakes, pastries, bread).
 - **Cake Flour / Pastry Flour (Soft Wheat):** Low in protein. Produces a fine, tender crumb.

- **OTHER FLOURS (NON-WHEAT):**

- **Corn Flour:** From maize (e.g., for "Tuwo Masara").
- **Rice Flour:** From rice (e.g., for rice cakes).
- **Soy Flour:** From soybeans (high protein, gluten-free).

2: LOCAL FLOUR SOURCES (NIGERIAN STAPLES)

Many of our starchy staples (Week 10, Term 1) are processed into flour. These are "gluten-free."

1. CASSAVA FLOUR:

- **Gari:** (Technically a "grits," not a fine flour).
- **Lafun:** A fine, white, fermented cassava flour.
- **Fufu:** (A wet paste, but also a flour product).

2. YAM FLOUR:

- **Amala (Elubo):** Made from dried, milled yam (yam flour). It has a characteristic dark colour and earthy flavour.

3. PLANTAIN FLOUR:

- Made from dried, milled *unripe* plantain. Also used to make "Amala."

4. BEAN FLOUR:

- Made from dried, milled black-eyed beans. Used as a "convenience" product to make *Akara* or *Moi-Moi* without having to soak and blend the beans.

3: NUTRITIVE VALUE OF FLOUR

- **CARBOHYDRATES (STARCH):** This is the primary nutrient (70-80%). Flour is a major source of energy.
- **PROTEIN:**
 - **Wheat flour:** Contains a significant amount of protein (gluten).
 - **Local flours:** (Yam, cassava) are very *low* in protein.
 - **Bean flour:** Is very *high* in protein.
- **FAT:** Very low in fat (unless it's a "nut flour" like almond flour).
- **VITAMINS/MINERALS:**
 - Contains some B-vitamins and iron.
 - **"FORTIFIED" FLOUR:** In many countries (including Nigeria), industrial flour is "fortified," meaning key nutrients (like Iron, Folic Acid, Vitamin B) are *added back* after milling. This is a crucial public health policy.

4: DISHES FROM FLOUR

- **NIGERIAN DISHES:**
 - **SWALLOWS (FUFU):** Eba (from Gari), Amala (from yam/plantain flour), Lafun.
 - **SNACKS:** Akara/Moi-Moi (from bean flour), Mosa (from plantain flour).
- **BAKED GOODS (MAINLY WHEAT FLOUR):**
 - **BREAD:** (Yeast-leavened).
 - **PASTRIES:** (High fat, flaky texture).
 - **Shortcrust:** (e.g., Meat pie, tart shells).
 - **Puff Pastry:** (e.g., sausage rolls, croissants).
 - **Choux Pastry:** (e.g., Eclairs, puff-puffs).
 - **CAKES:** (Chemically-leavened). (e.g., Queen cakes, sponges).
 - **BISCUITS (COOKIES):** (e.g., shortbread).

- **OTHER:**

- **THICKENER:** Flour is used to thicken stews, soups, and gravies.

EVALUATION

1. What is "flour," and what is "gluten"?
2. Why is "gluten" the most important protein in baking bread?
3. What is "Amala," and what two local ingredients can be milled to make it?
4. What is "fortified flour"?
5. List three types of "pastries" and a dish made from one of them.

ASSIGNMENT

1. Write a 400-word essay "Compare and Contrast" the properties of "Bread Flour" (hard wheat) and "Cake Flour" (soft wheat).
2. What is "Celiac Disease"? Research this condition and explain why a person with it cannot eat wheat flour.
3. "Eba" and "Fufu" are both from cassava. Research and describe the difference in *processing* that results in these two different products.
4. Develop a recipe for a simple "Shortcrust Pastry" (e.g., for meat pies), listing the 4-5 basic ingredients and the method.
5. What is the "convenience" advantage of using "Bean Flour" instead of whole beans?

WEEK 10: RAISING AGENTS

TOPIC: RAISING AGENTS IN FLOUR COOKERY

INTRODUCTION: A "raising agent" (or "leavening agent") is any substance that produces gas in a dough or batter. This gas gets trapped (often by gluten or an egg structure), causing the mixture to expand and rise, resulting in a light, soft, porous texture. Without raising agents, all baked goods would be flat, hard, and dense.



1: HOW RAISING AGENTS WORK

1. **GAS IS PRODUCED:** The agent (e.g., yeast, baking powder) creates gas (usually carbon dioxide or air).
2. **GAS IS TRAPPED:** The "structure" of the batter (the gluten network in bread, the egg proteins in a cake) traps the gas bubbles.
3. **GAS EXPANDS:** When heated in the oven, the trapped gas bubbles expand, pushing the mixture upwards.
4. **MIXTURE SETS:** The heat eventually "sets" (coagulates) the proteins and starches, permanently locking the airy, porous structure in place.

2: TYPES OF RAISING AGENTS

- **1. AIR (MECHANICAL):**
 - **HOW IT WORKS:** Air is physically "beaten" into a mixture.
 - **METHODS:**
 - **CREAMING:** Beating butter and sugar together until light and fluffy (traps air). Used for cakes.
 - **WHISKING:** Whisking egg whites (for meringues) or whole eggs (for sponges) traps large amounts of air.
 - **SIEVING:** Sieving flour traps a small amount of air.
 - **FOLDING:** Gently folding in flour to *keep* the trapped air.
- **2. STEAM (NATURAL):**
 - **HOW IT WORKS:** All batters contain liquid (water, milk). When heated, this liquid turns to steam. Steam expands to *1,600 times* its original volume, pushing the mixture up.
 - **USED IN:**
 - **Puff Pastry:** The "puff" is from layers of butter and dough creating steam.
 - **Choux Pastry:** (Puff-puffs) The high moisture content turns to steam, creating a hollow centre.
 - **Yorkshire Puddings.**
- **3. BIOLOGICAL (YEAST):**
 - **WHAT IS IT:** Yeast is a *living, single-celled micro-organism*.
 - **HOW IT WORKS (FERMENTATION):**

- Yeast "eats" sugar (in the flour/dough).
 - **CONDITIONS:** It needs **Warmth** (not too hot), **Moisture** (water), and **Food** (sugar) to activate.
 - **BY-PRODUCTS:** As it feeds, it produces **Carbon Dioxide (CO₂)** gas and **Alcohol** (which burns off in baking).
- **USED IN:** Bread, Doughnuts.
- **PROS:** Gives a unique, rich flavour.
- **CONS:** It is *slow* (dough must be left to "prove" or rise for 1-2 hours).
- **4. CHEMICAL:**
 - **HOW IT WORKS:** Two chemicals (an acid and an alkali) react *when wet* to produce CO₂ gas.
 - **A. BAKING SODA (BICARBONATE OF SODA / ALKALI):**
 - **COMPOSITION:** 100% Sodium Bicarbonate (an alkali).
 - **HOW IT WORKS:** It *must* be mixed with an **acidic ingredient** (e.g., buttermilk, yogurt, lemon juice, vinegar, brown sugar) to react and produce gas.
 - **PROBLEM:** If you don't add enough acid, it leaves a "soapy" taste.
 - **B. BAKING POWDER (A "COMPLETE" MIXTURE):**
 - **COMPOSITION:** This is the "convenience" product. It is a pre-mixed blend of:
 1. Baking Soda (Alkali)
 2. An Acid (e.g., Cream of Tartar)
 3. A Starch (e.g., cornflour) to keep it dry and stop it from reacting.
 - **HOW IT WORKS:** It already *contains* the acid and alkali. You just add liquid (water, milk) to start the reaction.
 - **"DOUBLE-ACTING":** Most modern baking powders react *twice*: once when wet (in the bowl) and again when heated (in the oven).
 - **USED IN:** Cakes, biscuits, cookies, scones, quick breads.

EVALUATION

1. What is a "raising agent," and what is its primary purpose in baking?

2. List the four main *types* of raising agents.
3. What are the three conditions that *yeast* needs to activate?
4. What gas does yeast produce through "fermentation"?
5. What is the key difference between "Baking Soda" and "Baking Powder"? Which one *requires* you to add an acidic ingredient?

ASSIGNMENT

1. Write a 400-word report on "Yeast." Explain the fermentation process in detail (sugar → CO₂ + alcohol) and its role in bread-making.
2. What is "Creaming"? Explain the *science* of creaming butter and sugar and how it acts as a raising agent.
3. You have a recipe that asks for 1 cup of buttermilk and 1 tsp of baking soda. You only have fresh milk. How could you "make" your own buttermilk? (Hint: involves an acid).
4. Design an experiment to test the effectiveness of three different raising agents. (e.g., "Dough A: Yeast," "Dough B: Baking Powder," "Dough C: No agent"). Predict the results.
5. What is "Sourdough Bread"? Research how it is leavened. (Hint: It uses a "wild" yeast starter).

WEEK 11: REVISION

TOPIC: GENERAL REVIEW OF ALL TOPICS

INTRODUCTION: This week is dedicated to a comprehensive review of all topics covered in SS2 Second Term. We will consolidate our knowledge and prepare for the final examinations, focusing on management, equipment, nutrition, and baking.

REVISION SCOPE:

- **KITCHEN & ENERGY MANAGEMENT:**
 - **Energy Management:** Principles (work simplification, dovetailing), fatigue.
 - **Equipment:** Classification, selection, care (wood, non-stick, motor base).
 - **Labour Savers:** Blenders, pressure cookers, etc.
- **NUTRITION & MEAL PLANNING:**
 - **Lifecycle Needs:** Detailed needs for Infants, Children, Adolescents, Adults, Pregnant/Lactating, and Elderly.
 - **Special Groups:** Invalids, Vegetarians (Lacto-ovo, Vegan), Athletes.
 - **Key Nutrients:** Iron, Calcium, Folate, B12, Complementary Proteins.
- **FLOUR COOKERY:**
 - **Flour:** Gluten, types (hard/soft), local flours (yam, cassava).
 - **Raising Agents:** Air (creaming/whisking), Steam, Yeast (fermentation), Chemical (Baking Soda vs. Powder).
- **EXAM PREPARATION:**
 - **Instructional Verbs:** List vs. Explain vs. Discuss.

REVISION STRATEGY:

1. **Review all notes.**
2. **Answer all "Evaluation" questions** from each week.
3. **PRACTICE ESSAY QUESTIONS:**
 - "Discuss the principles of energy management (work simplification) that can be applied to reduce fatigue in the kitchen."

- "Compare and contrast the nutritional needs of an adolescent boy and an 80-year-old man."
- "A family friend is pregnant. Advise her on her nutritional needs, focusing on 3 key nutrients and 3 foods to avoid."
- "Explain the four types of raising agents, giving a clear example of how *each one* works and a food product it is used in."
- "You are buying a new electric blender. Discuss five factors that will guide your selection."



SS2 FOOD AND NUTRITION LESSON NOTES

THIRD TERM

WEEK 1: CONVENIENCE FOODS

TOPIC: CONVENIENCE FOODS

INTRODUCTION: A "convenience food" is any food product that has been partially or fully prepared by a manufacturer, requiring minimal further preparation (or "convenience") from the consumer. These foods are a defining feature of the modern diet, offering benefits in time-saving but also posing nutritional and financial challenges.



1: MEANING, USES, FORMS, AND EXAMPLES

- **MEANING:** Convenience foods are food items that have been processed to reduce the amount of time and effort (labour) required by the consumer for their preparation. The "convenience" is what is being sold.

- **USES:**

- To save time in busy households.
- To provide quick meals when cooking skills are limited.
- Used as an emergency or "stand-by" food.
- To add variety to meals quickly (e.g., using a pre-made sauce).

- **FORMS AND EXAMPLES:**

1. **CANNED FOODS:** (Preserved in cans).

- *Examples:* Canned sardines, corned beef, baked beans, evaporated milk, tomato purée (paste), canned sweetcorn.

2. **FROZEN FOODS:** (Preserved by freezing).

- *Examples:* Frozen chicken/turkey, frozen fish (e.g., "carton fish"), frozen peas, ice cream.

3. **DRIED (DEHYDRATED) FOODS:** (Preserved by removing water).

- *Examples:* Powdered milk, instant coffee, stock cubes (e.g., Maggi, Knorr), instant noodles (e.g., Indomie), bean flour, yam flour (Amala), breakfast cereals, Gari, pasta.

4. **READY-TO-EAT (RTE):** (Fully cooked, just need opening).

- *Examples:* Bread, biscuits, yogurt, meat pies, sausage rolls, packaged snacks (e.g., plantain chips).

5. **READY-TO-COOK / "JUST ADD" MIXES:** (Requires one or two simple steps).

- *Examples:* Cake mixes, custard powder, "Jollof rice" spice mixes.

2: FACTORS TO CONSIDER IN SELECTION

When choosing convenience foods, it is important to be a smart consumer.

1. **NUTRITIONAL VALUE (READ THE LABEL):** This is the most important factor.

- **SODIUM (SALT):** Many (especially canned and instant) are extremely high in salt.
- **SUGAR:** Check for "hidden" sugars in cereals, sauces, and drinks.

- **FAT (ESPECIALLY SATURATED/TRANS):** High in many pastries and frozen meals.
 - **FIBRE:** Often very low in processed foods.
2. **COST (VALUE FOR MONEY):**
- Is it cheaper to make from scratch? (e.t., a homemade sauce vs. a jar).
 - Is the convenience "worth" the extra price?
 - Compare cost per gram/kg, not just the unit price.
3. **QUALITY AND APPEARANCE OF PACKAGING:**
- **CANS: DO NOT BUY** if they are bulging, leaking, badly dented, or rusty. This can indicate *botulism* (a deadly food poisoning).
 - **PACKAGES:** Ensure seals are intact, not torn or open.
 - **FROZEN:** Check for signs of "freezer burn" or thawing (ice crystals).
4. **EXPIRATION DATE (BEST BEFORE / USE BY):**
- **"Use By":** Relates to safety. Do not eat after this date.
 - **"Best Before":** Relates to quality. It may be safe but not as good.
5. **STORAGE:** Do you have the correct storage for it? (e.g., Don't buy frozen food if you have no freezer).
6. **INGREDIENTS LIST:** Check for allergens (e.g., nuts, soy) or additives you wish to avoid (e.g., MSG, artificial colours).

3: ADVANTAGES AND DISADVANTAGES

- **ADVANTAGES:**
 1. **SAVES TIME AND LABOUR:** This is the primary benefit. (e.g., using bean flour for Akara).
 2. **REDUCES PREPARATION & CLEANUP:** Less chopping, peeling, and washing up.
 3. **AVAILABILITY:** Allows consumption of non-seasonal foods (e.g., canned sweetcorn in any season).
 4. **LONG SHELF LIFE:** Can be stored for long periods, which is good for emergencies (e.g., canned fish, gari).
 5. **VARIETY:** Can add variety to the diet easily.

6. **LIGHTWEIGHT & PORTABLE:** (e.g., instant noodles are good for students, travellers).
- **DISADVANTAGES:**
 1. **POOR NUTRITIONAL VALUE:** Often very high in salt (sodium), sugar, and unhealthy fats.
 2. **LOW IN FIBRE:** Most processing removes dietary fibre.
 3. **HIGH COST:** Often more expensive than making the equivalent dish from scratch.
 4. **LOSS OF NUTRIENTS:** The processing (especially high-heat canning) can destroy heat-sensitive vitamins (like Vitamin C).
 5. **ARTIFICIAL ADDITIVES:** May contain high levels of preservatives, artificial colours, and flavour enhancers.
 6. **"PALATE FATIGUE":** Many processed foods have a similar "factory" taste.

4: DISHES MADE WITH CONVENIENCE FOODS

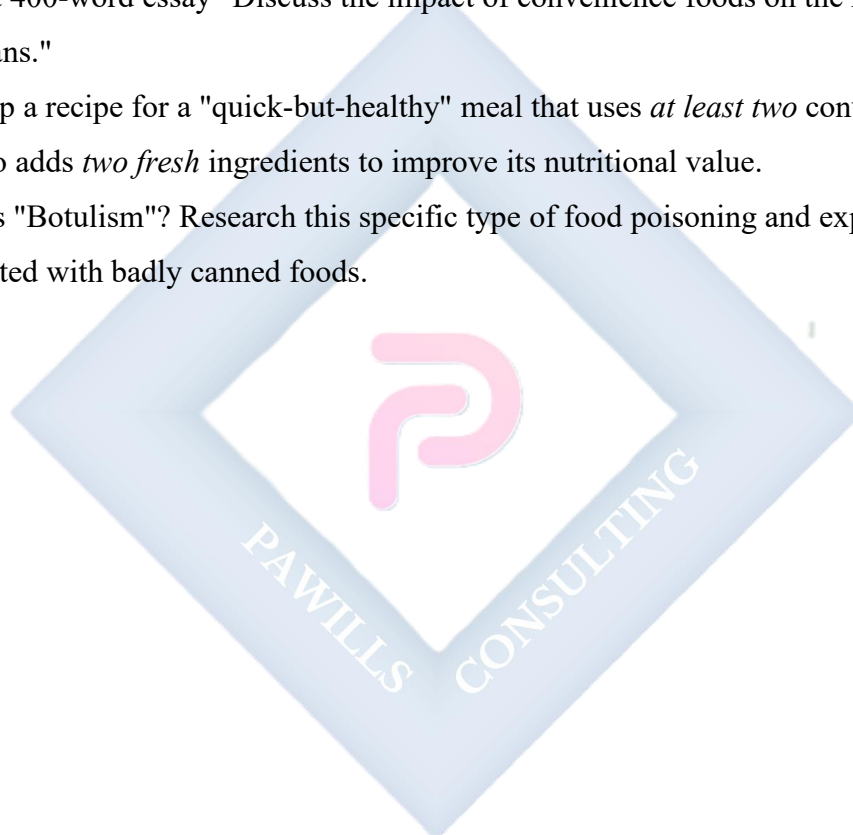
- **INSTANT NOODLES:** Cooked with egg, vegetables, or canned sardines.
- **BREAD:** Eaten with canned baked beans, corned beef, or as a sandwich.
- **GARI (EBA):** Eaten with a stew (which may be made from canned tomatoes).
- **CUSTARD:** Made from custard powder and evaporated milk, served with akara (from bean flour).
- **"QUICK" JOLLOF RICE:** Using a pre-made "Jollof" spice mix.
- **SPAGHETTI AND CORNED BEEF SAUCE:** Using dried pasta and canned meat.

EVALUATION

1. What is a "convenience food"?
2. List the four main "forms" of convenience foods and give one example for each.
3. What are the three things you must **never** see on a can of food you are about to buy? (e.g., bulging).
4. List three major *advantages* of convenience foods.
5. List three major *disadvantages* (nutritional or otherwise) of convenience foods.

ASSIGNMENT

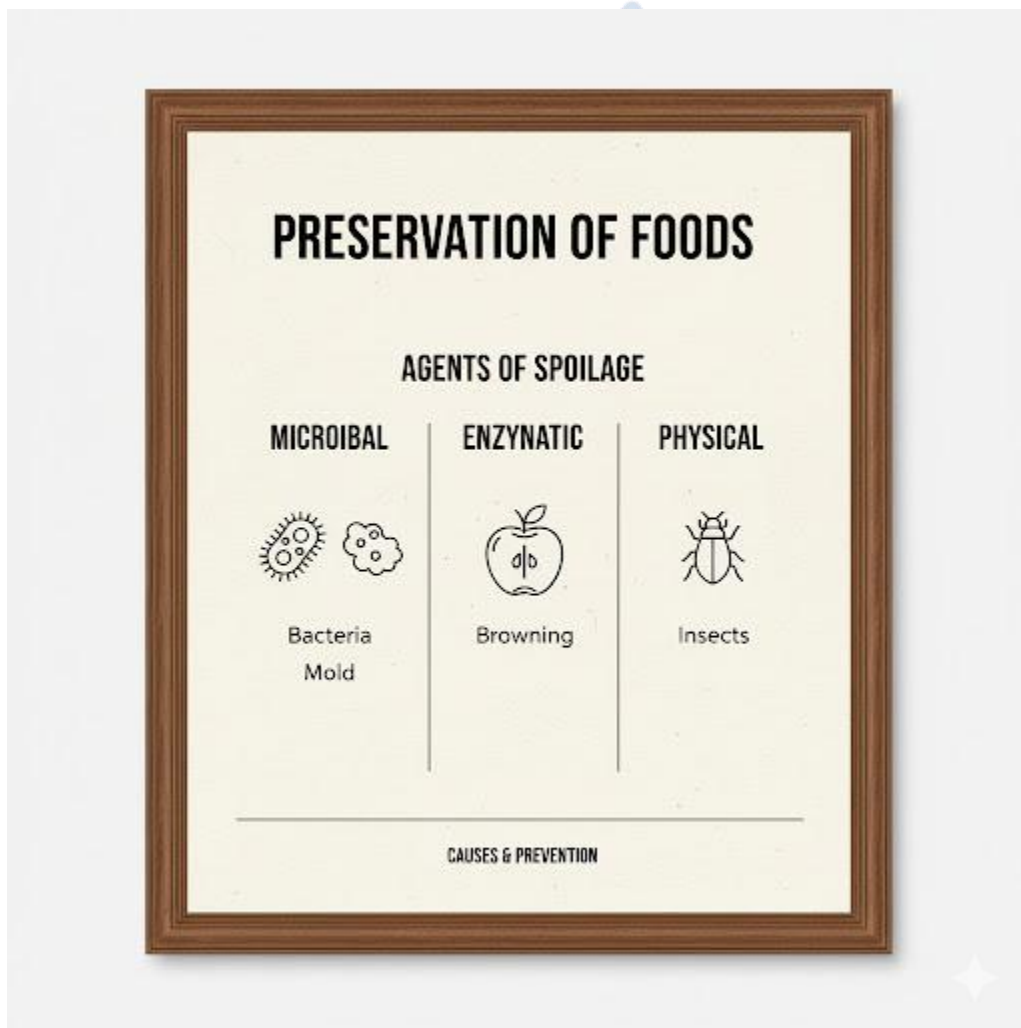
1. Go to your kitchen cupboard. Identify five different convenience foods. Create a table with these columns: (1) Product Name, (2) Form (e.g., Canned, Dried), (3) "Convenience" (What task does it save?).
2. Select one of the convenience foods from Q1 (e.g., Indomie, a stock cube). Find its nutritional label. Write down the (a) Sodium content, (b) Sugar content, and (c) Fat content per 100g. Write a paragraph on whether you think it's a "healthy" choice.
3. Write a 400-word essay "Discuss the impact of convenience foods on the health of urban Nigerians."
4. Develop a recipe for a "quick-but-healthy" meal that uses *at least two* convenience foods but also adds *two fresh* ingredients to improve its nutritional value.
5. What is "Botulism"? Research this specific type of food poisoning and explain why it is associated with badly canned foods.



WEEK 2: PRESERVATION OF FOODS

TOPIC: FOOD SPOILAGE AND PRINCIPLES OF PRESERVATION

INTRODUCTION: From the moment food is harvested or slaughtered, it begins to spoil. Food preservation is the science of treating and handling food to stop or slow down this spoilage, loss of quality, and inedibility. It is one of the oldest and most important aspects of food science, allowing us to store food for emergencies, for transport, or for future use.



1: MEANING OF FOOD SPOILAGE

- **MEANING:** Food spoilage is a process that causes food to degrade and become undesirable or unsafe for consumption. It is any change that renders the food offensive to our senses (taste, smell, sight, touch).
- **SIGNS OF SPOILAGE:**
 - **ODOUR:** "Off" smells, sourness (e.g., sour milk), putrid smells (e.g., rotting meat).
 - **SIGHT:** Mould growth (blue/green/white fuzz on bread or tomatoes), discolouration, slime.
 - **TOUCH:** Slimy texture (on meat), mushiness (on vegetables).
 - **TASTE:** Sour, bitter, or "off" flavours (Note: You should **never** "taste-test" food you suspect is spoiled).

2: CAUSES AND AGENTS OF FOOD SPOILAGE

Food spoilage is not magic; it is a chemical and biological process caused by specific agents.

1. MICROBIAL (THE MAIN CAUSE):

- **MEANING:** Spoilage caused by microscopic living organisms.
- **AGENTS:**
 - **BACTERIA:** (e.g., *Salmonella*, *E. coli*). Cause food poisoning, rot, and slime.
 - **MOULDS:** (e.g., on bread, citrus, tomatoes). Cause "furry" growth. Some produce dangerous *mycotoxins* (e.g., aflatoxin on groundnuts).
 - **YEASTS:** (e.g., on fruit juice, palm wine). Cause *fermentation*, leading to a sour, alcoholic taste and smell.

2. CHEMICAL (ENZYMATIC):

- **MEANING:** Spoilage caused by *enzymes* that are naturally present *in the food itself*.
- **AGENTS (ENZYMES):** These are biological catalysts that speed up chemical reactions.

- **OXIDATION & BROWNING:** When you cut an apple or yam, enzymes (like PPO) react with oxygen in the air, causing it to turn brown. This is an enzymatic reaction.
- **RIPENING:** Enzymes are responsible for ripening (e.g., turning a banana from starchy to sweet). If left unchecked, these same enzymes will cause *over-ripening* (mushiness, decay).

3. PHYSICAL:

- **MEANING:** Spoilage or loss of quality caused by physical damage or environmental factors.
- **AGENTS:**
 - **BRUISING/DAMAGE:** (e.g., a bruised tomato) breaks the skin, allowing microbes to enter.
 - **INSECTS & RODENTS:** (e.g., weevils in beans, rats on gari). They damage, eat, and contaminate food with their waste.
 - **LIGHT:** (e.g., sunlight) can destroy vitamins (like Riboflavin) in milk.
 - **MOISTURE/HUMIDITY:** Causes dry foods (like gari or biscuits) to become soft and caked.
 - **DRYNESS:** Causes moist foods (like vegetables) to wilt.

3: IMPORTANCE OF FOOD PRESERVATION

Why do we bother preserving food?

1. **INCREASES SHELF LIFE:** This is the primary goal. It allows food to be stored for days, months, or years.
2. **PREVENTS FOOD WASTAGE:** Allows us to store surplus food from a good harvest, preventing it from rotting.
3. **ENSURES FOOD SECURITY:** Allows families and nations to store food for the "lean season" (when harvests are poor) or emergencies (e.g., war, famine).
4. **SAVES MONEY:** You can buy food in bulk when it is cheap (in season) and preserve it for when it is expensive (out of season).

5. **INCREASES VARIETY:** Allows you to eat non-seasonal foods (e.g., eating dried mango in the rainy season).
6. **AIDS TRANSPORTATION:** Allows perishable foods (like milk or fish) to be transported long distances (e.g., as powdered milk or smoked fish).
7. **CREATES NEW PRODUCTS:** Some preservation methods create entirely new, desirable foods (e.g., fermentation turns milk into yogurt and cheese).

4: PRINCIPLES OF FOOD PRESERVATION

All preservation methods work by targeting the *agents of spoilage*. They are based on one or more of these scientific principles:

1. INACTIVATION/KILLING OF MICROBES:

- **PRINCIPLE:** Most microbes (bacteria, yeast, mould) are killed by high heat.
- **METHODS:** Boiling, Frying, Canning, Pasteurization.

2. REMOVAL OF WATER (MOISTURE):

- **PRINCIPLE:** All microbes need water (moisture) to survive and multiply. If you remove the water, they become dormant or die.
- **METHODS:** Drying (sun-drying, dehydration), Salting (salt draws water *out* of food), Sugaring (sugar draws water *out*).

3. REMOVAL OF OXYGEN (AIR):

- **PRINCIPLE:** Many microbes (and enzymes) need oxygen to function (they are "aerobic").
- **METHODS:** Canning, Vacuum packing, Bottling (e.g., covering with oil).

4. SLOWING DOWN/INHIBITING GROWTH (NOT KILLING):

- **PRINCIPLE:** Slowing down microbial growth and enzymatic reactions.
- **METHODS:**
 - **COLD (REFRIGERATION/FREEZING):** Low temperatures make microbes and enzymes *dormant* (they are not killed).
 - **CHANGING PH (ACIDITY):** Most microbes cannot grow in a highly acidic environment.

- **METHODS:** Pickling (using vinegar - acetic acid), Fermentation (bacteria create lactic acid).

5. INACTIVATION OF ENZYMES:

- **PRINCIPLE:** Enzymes (which cause browning/ripening) are destroyed by heat.
- **METHODS: Blanching** (a quick dip in boiling water) is done to vegetables before freezing to destroy enzymes.

EVALUATION

1. What is "food spoilage"? List three signs of spoilage.
2. Name the three main *causes* of food spoilage (microbial, chemical, physical).
3. What are the three types of *microbial* agents? Which one causes fermentation?
4. What is an *enzyme*, and what role does it play in the browning of a cut yam?
5. List four different *scientific principles* of food preservation.

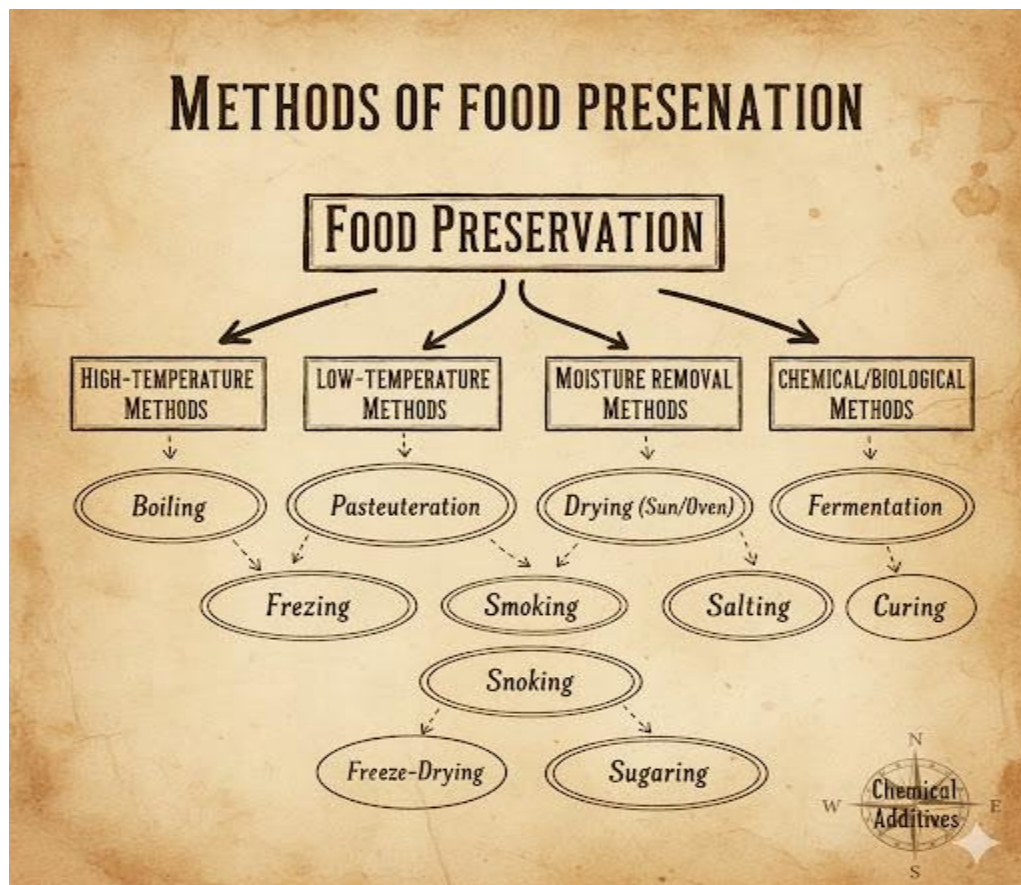
ASSIGNMENT

1. Take a single slice of bread. Place it in a plastic bag with a *few drops* of water, seal it, and leave it in a warm, dark place for 3-5 days.
 - (a) Observe it each day and write down what you see.
 - (b) What agent of spoilage are you observing?
 - (c) What *conditions* did you provide to encourage its growth?
2. What is an "enzyme"? Write a 400-word essay on the role of enzymes in food, explaining how they can be both *good* (e.g., in ripening) and *bad* (e.g., in browning and over-ripening).
3. Research "Aflatoxins." What are they, what mould produces them, and what common Nigerian staple food is at high risk?
4. What is "blanching"? Explain the step-by-step process and why it is a *necessary* first step before *freezing* most vegetables.
5. Explain the *two* ways that "salting" preserves fish. (Hint: one is about water, one is about bacteria).

WEEK 3: METHODS OF FOOD PRESERVATION

TOPIC: METHODS OF FOOD PRESERVATION

INTRODUCTION: Based on the principles we learned last week, humanity has developed many methods, from ancient to modern, to preserve food. This week, we will explore the "how-to" of these methods, classifying them by the principles they use.



1: HIGH TEMPERATURE METHODS (KILLING MICROBES)

- **1. BOILING / COOKING:**
 - **HOW IT WORKS:** Heats food to 100°C, which kills most active bacteria and destroys enzymes.

- **USE:** A temporary, short-term method. (e.g., Boiling meat for a stew, boiling milk to use the next day).
- **2. PASTEURIZATION:**
 - **HOW IT WORKS:** Heats liquid (like milk) to a high temperature (e.g., 72°C) for a short time (15 sec) to kill *pathogenic* (harmful) bacteria. It does *not* kill all microbes, so the food must be refrigerated.
 - **USE:** Milk, fruit juices.
- **3. CANNING / BOTTLING:**
 - **HOW IT WORKS:** Food is packed into an airtight container (can or jar) and heated to a very high temperature to kill *all* microbes and enzymes. As it cools, a vacuum seal forms, preventing re-contamination and removing oxygen.
 - **PRINCIPLES USED:** High heat (kills microbes), Removal of oxygen, Airtight seal (prevents re-contamination).
 - **USE:** Fish, meat, vegetables (e.g., tomatoes), fruits.
- **4. FRYING:** High-heat (150°C+) oil dehydrates the surface and kills microbes. Often used with other methods.

2: LOW TEMPERATURE METHODS (SLOWING GROWTH)

- **1. REFRIGERATION (CHILLING):**
 - **HOW IT WORKS:** Storing food at a low temperature (0-4°C). This *slows down* (but does not stop) the growth of microbes and the action of enzymes.
 - **USE:** A short-term method for perishable foods like milk, fresh meat, and leftovers.
- **2. FREEZING:**
 - **HOW IT WORKS:** Storing food at -18°C or below. This *stops* microbial growth completely (makes them dormant) and locks water as ice.
 - **USE:** A long-term method for meat, fish, poultry, vegetables (after blanching), and even pre-cooked meals.
 - **NOTE:** It does *not* kill bacteria; they "wake up" when the food is thawed.

3: MOISTURE REMOVAL METHODS (REMOVING WATER)

- **1. SUN-DRYING:**

- **HOW IT WORKS:** An ancient, traditional method. Food is spread in the open air to be dehydrated by the sun and wind.
- **USE:** Pepper, okra (for powder), fish (stockfish), cassava (for flour).
- **CONS:** Slow, dependent on weather, and at high risk of contamination by insects, dust, and rodents.

- **2. DEHYDRATION (MODERN):**

- **HOW IT WORKS:** Using a machine (a "dehydrator") or an oven on low heat to circulate hot, dry air.
- **USE:** Making beef jerky, dried fruits, herbs. More hygienic and faster.

- **3. SALTING:**

- **HOW IT WORKS:** Salt is a "hydrophilic" (water-loving) substance.
 1. It draws water *out* of the food (dehydration).
 2. It draws water *out* of any bacteria (osmosis), killing them.
 3. It is toxic to many microbes.
- **USE:** Preserving fish, meat, and vegetables.

- **4. SUGARING:**

- **HOW IT WORKS:** Works exactly like salt (osmosis). A high concentration of sugar (e.g., in jam) draws water out of microbes.
- **USE:** Jams, jellies, condensed milk, candied fruits.

- **5. SMOKING:**

- **HOW IT WORKS:**
 1. The hot air from the fire *dries* the food.
 2. The chemicals in the wood smoke (e.g., phenols) are *anti-bacterial* and are deposited on the food.
- **USE:** Fish (smoked fish), meat (e.g., Asun, jerky).

4: CHEMICAL / BIOLOGICAL METHODS

- **1. PICKLING (USING ACID):**

- **HOW IT WORKS:** Involves soaking food in a high-acid liquid (usually vinegar - acetic acid). The low pH (high acidity) prevents most microbes from growing.

- **USE:** Vegetables (onions, cucumbers, peppers).
- **2. FERMENTATION:**
 - **HOW IT WORKS:** An ancient "probiotic" method. Uses "good" microbes (yeast or bacteria) to convert sugars into acid or alcohol. This *acid* or *alcohol* then acts as the preservative.
 - **EXAMPLES:**
 - **LACTIC ACID:** (e.g., Gari/Fufu - fermentation of cassava creates acid), (e.g., Yogurt - bacteria turn lactose into lactic acid).
 - **ALCOHOL:** (e.g., Palm wine - yeast turns sugar into alcohol).
- **3. USE OF CHEMICAL PRESERVATIVES:**
 - **HOW IT WORKS:** Adding chemicals that are anti-microbial or anti-oxidant.
 - **EXAMPLES:** Benzoates (in soft drinks), Sorbates (in bread), Sulphur dioxide (in dried fruits to keep colour).

EVALUATION

1. What is the main difference in *principle* between "Canning" and "Refrigeration"? (Hint: what happens to the microbes?).
2. What is "blanching," and why must it be done before freezing vegetables? (Links to Week 2).
3. Explain the two preservative effects of "Smoking."
4. Explain the two preservative effects of "Salting."
5. What is "fermentation"? Name two Nigerian foods that are preserved using this method.

ASSIGNMENT

1. Choose *one* traditional Nigerian preservation method (e.g., sun-drying, smoking, making gari). Write a 400-word report describing the step-by-step process and the scientific principles at work.
2. What is the difference between "Pasteurization" and "Canning" (Sterilization)? Create a table comparing them (Temperature, Time, Effect on microbes, Shelf life, Storage).

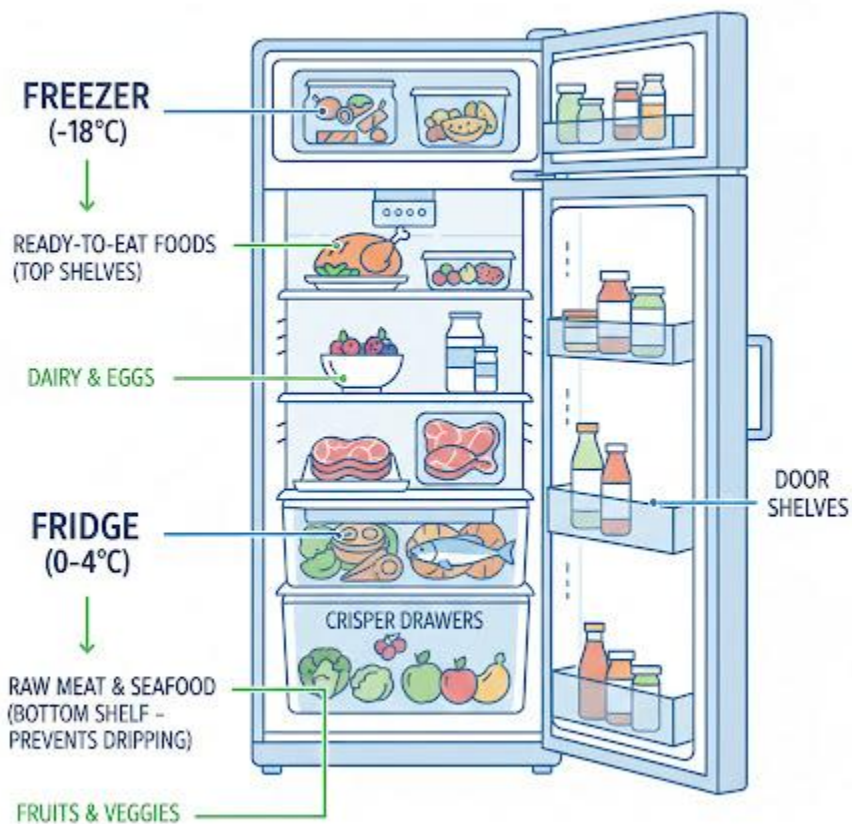
3. Why must you *not* re-freeze food (especially meat) that has already been thawed? (Hint: what happens to the bacteria?).
4. Design a simple home "sun-drying" rack for peppers or herbs that would protect the food from insects and dust. Draw a diagram and explain your design.
5. What is "jam"? Explain how *sugar* and *pectin* (a gelling agent) work together to preserve fruit.



WEEK 4: FOOD STORAGE

TOPIC: FOOD STORAGE

INTRODUCTION: Food preservation is useless if the food is not *stored* correctly. Food storage is the management of food in the home (or commercially) to maintain its quality and safety after purchasing or preserving. Proper storage prevents waste, saves money, and ensures food is safe to eat.



1: MEANING AND IMPORTANCE OF FOOD STORAGE

- **MEANING:** Food storage is the act of keeping food in a safe, clean, and appropriate environment to prevent spoilage and maintain nutritional value.

- **IMPORTANCE:**

- **PREVENTS FOOD WASTAGE:** The primary goal.
- **MAINTAINS FOOD SAFETY:** Prevents food poisoning by stopping the growth of harmful bacteria.
- **PRESERVES NUTRITIONAL VALUE:** Protects vitamins and minerals from (e.g.) light, air, and heat.
- **SAVES MONEY:** Stops you from throwing away spoiled food. Allows you to "bulk-buy" and save.
- **ENSURES HYGIENE:** Protects food from contamination by pests (rats, insects), dust, and pets.
- **CONVENIENCE:** A well-organized store (pantry, fridge) makes it easy to find ingredients.

2: STORAGE OF PERISHABLE FOODS

- **PERISHABLE FOODS:** These are moist, high-protein foods that spoil *very quickly* (within hours or days) at room temperature. They are high-risk.
- **EXAMPLES:** Fresh meat, poultry, fish, milk, eggs, most fresh vegetables and fruits, leftovers.
- **STORAGE METHOD: REFRIGERATOR (0-4°C):**
 - **PRINCIPLE:** Cold temperatures slow down microbial growth.
 - **RULES:**
 1. **"RAW BELOW COOKED":** This is the **most important rule**.
 - Store raw meat/fish/poultry on the *bottom shelf*.
 - Store cooked/ready-to-eat foods (e.g., leftovers, milk, yogurt) on the *top shelves*.
 - **REASON:** To prevent raw meat juices (which contain bacteria) from *dripping onto* and *contaminating* cooked food. This is a major cause of food poisoning.
 2. **COVER ALL FOOD:** Cover leftovers with a lid or cling film to prevent them from drying out and absorbing odours.

3. **COOL HOT FOOD FIRST:** Do not put a large, steaming pot of stew *directly* into the fridge. It will raise the internal temperature of the fridge, putting *all* other food at risk. Cool it on the counter first.
4. **DO NOT OVERLOAD:** Do not pack the fridge so tightly that air cannot circulate.

3: STORAGE OF NON-PERISHABLE FOODS

- **NON-PERISHABLE (DRY) FOODS:** These are foods with low moisture content that have a long shelf life (months or years) *if* stored correctly.
- **EXAMPLES:** Rice, beans, gari, yam flour, pasta, sugar, salt, dried/canned/bottled foods.
- **STORAGE METHOD: PANTRY / LARDER (COOL, DARK, DRY):**
 - **PRINCIPLE:** Keep them dry, keep them cool, and keep them away from pests.
 - **RULES:**
 1. **AIRTIGHT CONTAINERS:** This is the best practice. Store rice, beans, flour, etc., in plastic, metal, or glass containers with tight-fitting lids.
 - **REASON:** Protects from humidity (moisture) and *all* pests (insects, rodents).
 2. **COOL, DARK PLACE:** Store in a pantry, cupboard, or "store." Do not store them next to the stove (heat) or in direct sunlight (light).
 3. **FIFO (FIRST IN, FIRST OUT):** When you buy new stock, put it at the *back* of the shelf. Use the *older* stock at the front first. This ensures food is rotated and doesn't expire.
 4. **CHECK CANS:** Check your canned goods for any bulging or rust.

4: STORAGE EQUIPMENT AND SPACES

- **1. REFRIGERATOR:**
 - For short-term storage of perishable foods.
 - **COMPONENTS:**
 - **Fridge Compartment (0-4°C):** For chilling.
 - **Freezer Compartment (-18°C):** For long-term frozen storage.

- **Crisper Drawer:** A drawer at the bottom designed to hold *humidity*, which keeps vegetables and fruits fresh and "crisp."
- **2. DEEP FREEZER / CHEST FREEZER:**
 - A standalone freezer for *long-term* storage (-18°C or below).
 - Ideal for storing bulk-purchased meat, fish, or pre-cooked meals.
- **3. PANTRY / LARDER (STORE):**
 - A cupboard or small room dedicated to storing dry goods, cans, and bottles.
 - **Ideal features:** Cool, dry, dark, well-ventilated, and has shelves.
- **4. BARN (TRADITIONAL / AGRICULTURAL):**
 - A large, well-ventilated structure used on farms to store *bulk* harvest.
 - **"Yam Barn":** A traditional West African structure with racks designed to store individual yam tubers, allowing air to circulate and prevent rot.
- **5. OTHER:**
 - **Racks / Shelves:** For yams, onions, potatoes (allows air circulation).
 - **Airtight Jars/Bins:** For gari, rice, beans.

EVALUATION

1. What is the primary *importance* of good food storage?
2. What is a "perishable" food? Give three examples.
3. What is the **most important rule** for organizing a refrigerator, and what is the *reason* for this rule?
4. What does the acronym "FIFO" stand for, and how is it used in food storage?
5. What is a "Yam Barn," and what is the principle behind its design?

ASSIGNMENT

1. Draw a simple, clear diagram of a refrigerator. Label the shelves (top, middle, bottom) and the crisper drawer. Using arrows, show where you would *correctly* store the following items: (a) raw chicken, (b) leftover stew, (c) milk, (d) carrots.
2. Write a 400-word essay on "The importance of Airtight Containers" for storing non-perishable foods. (Hint: discuss humidity, pests, and hygiene).

3. What is the "Danger Zone" for food safety? Research this concept (it's a temperature range) and explain why leftovers must be cooled and refrigerated quickly.
4. You buy a large bag of beans and find it has "weevils" (insects). Research and describe three *traditional* (non-chemical) methods for storing beans to prevent weevils.
5. Organize a small section of your home's food store (pantry or cupboard). Take a "before" and "after" photo (or draw a "before" and "after" diagram). Explain what you did and why (e.g., "I applied the FIFO principle...").



WEEK 5: MIDTERM TEST

TOPIC: ASSESSMENT OF WEEKS 1–4

INTRODUCTION: This week is dedicated to a formal assessment of all concepts covered in the first part of the term. The test will evaluate your understanding of convenience foods, spoilage, preservation, and storage.

SCOPE:

- **WEEK 1: CONVENIENCE FOODS:** Meaning, forms (canned, frozen, dried), advantages (time), disadvantages (salt, sugar, cost), selection (dented cans).
- **WEEK 2: FOOD PRESERVATION I:** Meaning of spoilage (signs), agents (microbes, enzymes, physical), importance of preservation (food security, waste reduction), principles (kill microbes, remove water/air, slow growth).
- **WEEK 3: FOOD PRESERVATION II:** Methods: High-temp (canning, pasteurization), Low-temp (freezing, refrigeration), Moisture removal (drying, salting, sugaring, smoking), Chemical/Bio (pickling, fermentation).
- **WEEK 4: FOOD STORAGE:** Importance, storage of perishables ("raw below cooked," refrigerator), storage of non-perishables (airtight, cool/dry, FIFO), storage equipment (freezer, barn, pantry).

FORMAT:

- **SECTION A: MULTIPLE CHOICE QUESTIONS (20 MARKS)** - Testing definitions and key facts.
- **SECTION B: SHORT ANSWER QUESTIONS (30 MARKS)** - Testing explanations (e.g., "Explain the 3 principles of food preservation," "List 3 signs of a bad can").
- **SECTION C: ESSAY/LONG ANSWER QUESTIONS (50 MARKS)** - (e.g., "Discuss the advantages and disadvantages of convenience foods," "Explain 5 different methods of food preservation, stating the scientific principle behind each one").

WEEK 6: PRACTICAL ON CONVENIENCE FOODS AND FOOD STORAGE

TOPIC: PRACTICAL PREPARATION AND DEMONSTRATION

INTRODUCTION: This week is a practical session. The objectives are:

1. To prepare and cook a balanced, appealing dish *using* convenience foods as a base, but *improving* their nutritional value with fresh ingredients.
2. To demonstrate a practical, traditional method of food preservation or a modern method of food storage.



1: PRACTICAL A - DISHES FROM CONVENIENCE FOODS

- **GOAL:** To show that "convenience" does not have to mean "unhealthy."

- **TASK:** Students will choose *one* of the following dishes to prepare. They must bring their own convenience food base.
- **SUGGESTED DISHES:**
 1. **"IMPROVED" INSTANT NOODLES:**
 - **CONVENIENCE:** 1 pack of instant noodles.
 - **IMPROVEMENTS:** Add fresh items like 1 boiled egg (protein), chopped carrots & green beans (vitamins/fibre), and a small amount of fresh pepper (flavour).
 2. **"QUICK" BEAN POTTAGE:**
 - **CONVENIENCE:** 1 can of cooked black-eyed beans OR 1 cup of bean flour.
 - **IMPROVEMENTS:** Fresh onion, palm oil, crayfish, and Ugu (leafy green) for iron and vitamins.
 3. **"EBA" AND SARDINE SAUCE:**
 - **CONVENIENCE:** Gari (for Eba), 1 can of sardines, canned tomato paste.
 - **IMPROVEMENTS:** Fresh onion, fresh tomatoes, and fresh pepper to make a proper sauce base for the sardine.
 4. **CUSTARD AND FRUIT:**
 - **CONVENIENCE:** Custard powder, powdered milk.
 - **IMPROVEMENTS:** Top the finished custard with fresh, sliced banana or pineapple (fibre, vitamins).

2: PRACTICAL B - DEMONSTRATION OF PRESERVATION/STORAGE

- **GOAL:** To demonstrate a preservation or storage principle learned in class.
- **TASK:** The teacher (or student groups) will demonstrate *one* of the following.
- **SUGGESTED DEMONSTRATIONS:**
 1. **PRESERVATION (PICKLING):**
 - **METHOD:** Thinly slice vegetables (onions, carrots, peppers). Make a hot brine (vinegar, water, salt, sugar). Pour over vegetables in a clean, airtight jar.
 - **PRINCIPLE:** Using acid (vinegar) to preserve.

2. PRESERVATION (BLANCHING FOR FREEZER):

- **METHOD:** Have a pot of boiling water and a bowl of ice water. Chop green beans. Drop in boiling water for 2 minutes, then *immediately* remove and plunge into ice water.
- **PRINCIPLE:** Inactivating enzymes before freezing.

3. STORAGE (ORGANIZING A "COOLER"):

- **METHOD:** Use a cooler box as a "mock fridge." Demonstrate the "raw below cooked" principle. Place (wrapped) raw chicken on the bottom and a (covered) bowl of cooked rice on the top shelf.
- **PRINCIPLE:** Correct, safe storage of perishables.

EVALUATION (PRACTICAL ASSESSMENT)

Students will be graded on:

1. **PLANNING (10%):** Quality of their plan for their chosen dish.
2. **HYGIENE & SAFETY (30%):** Clean work, correct handwashing, safe handling of equipment.
3. **SKILL & METHOD (40%):** Correct preparation, cooking, and *improvement* of the convenience food.
4. **FLAVOUR & PRESENTATION (20%):** The taste and appearance of the final dish.

ASSIGNMENT (POST-PRACTICAL)

1. Write a detailed "Post-Practical Report" on the dish you cooked in "Practical A."
2. The report must include: (a) The recipe, (b) The convenience food(s) used, (c) The *fresh* ingredients you added, and (d) *How* these fresh ingredients improved the nutritional value.
3. Why is it better to use *only half* the seasoning sachet in instant noodles and add fresh pepper/onion instead? (Hint: Sodium).
4. From "Practical B," explain the principle of *blanching* as if you were teaching an SS1 student.
5. From "Practical B," explain the principle of *pickling* as if you were teaching an SS1 student.



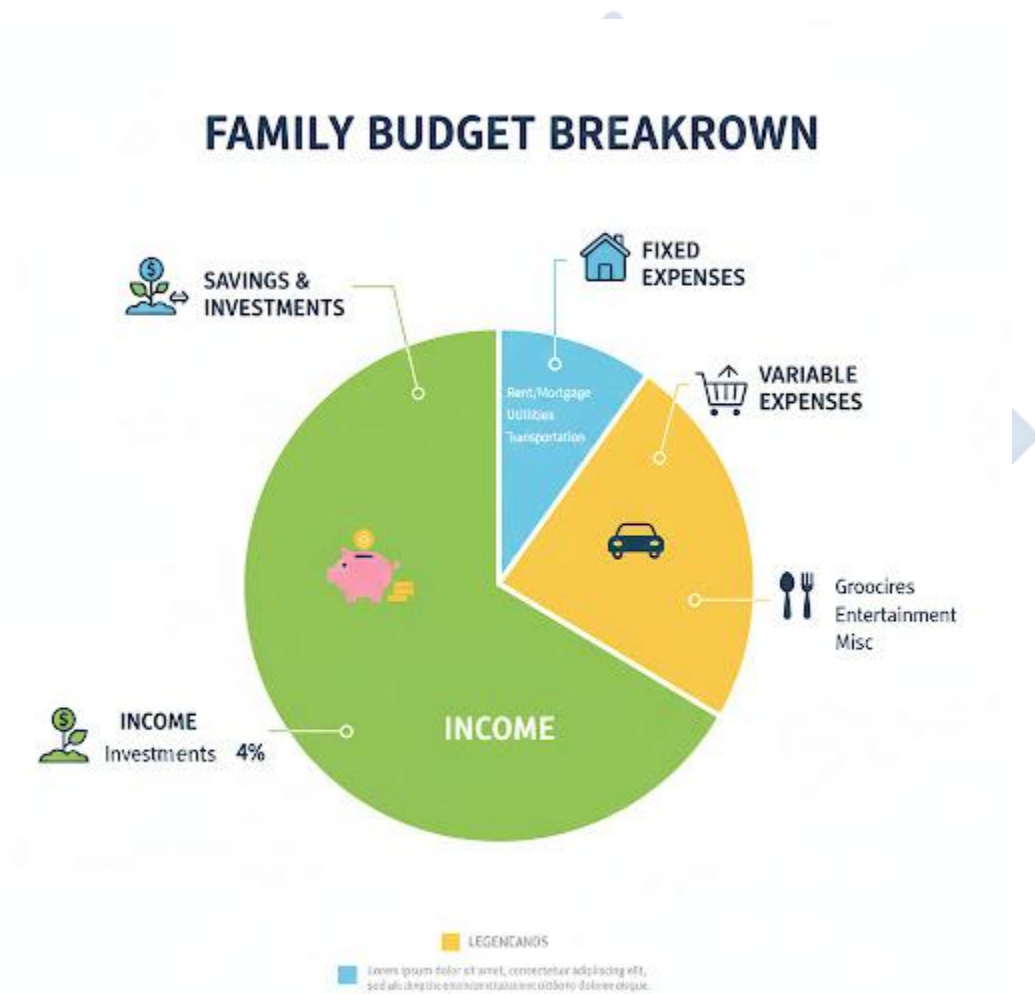
WEEK 7: MIDTERM BREAK



WEEK 8: BUDGETING

TOPIC: BUDGETING

INTRODUCTION: Budgeting is the cornerstone of all home management. A "budget" is simply a plan for how you will spend your money. It is a proactive and conscious process of balancing your "income" (money in) against your "expenditure" (money out) to achieve your financial goals.



1: MEANING OF KEY TERMS

- **BUDGET:** A detailed, written plan that estimates income and expenses for a specific period (e.g., a week or month).

- **INCOME:** All money *received* by a family.
 - **TYPES:** Salary, wages, profit from a business, "chop money" (allowance), gifts, rent from property.
- **EXPENDITURE:** All money *spent* by a family.
 - **TYPES:**
 - **FIXED EXPENSES:** Costs that are the *same* every month (e.g., rent, school fees, transport pass).
 - **VARIABLE EXPENSES (FLEXIBLE):** Costs that *change* based on usage (e.g., food, electricity bill, clothing, "NEPA").
 - **SAVINGS:** Money set aside for the future. **This should be treated as a "fixed expense" (Pay Yourself First).**
- **FAMILY NEEDS:** The items and services a family requires to live.
 - **NEEDS (ESSENTIAL):** Food, shelter (rent), clothing, healthcare, education.
 - **WANTS (NON-ESSENTIAL):** DSTV subscription, new phone, designer clothes, entertainment.

2: FOOD BUDGETING

- **MEANING:** A "food budget" is the specific portion of the family budget that is *allocated only for buying food*.
- **IMPORTANCE:**
 - Food is a *major* household expense (often the largest flexible expense).
 - It is a *variable* expense, meaning it is very easy to "overspend" on food if there is no plan.
 - A food budget ensures that the family eats *nutritiously* (allocating money for fruits/vegetables) and not just on "wants" (e.g., expensive snacks, takeaways).

3: FACTORS INFLUENCING BUDGETING

Why is one family's budget different from another's?

1. **FAMILY INCOME:** This is the most significant factor. You cannot (or should not) plan to spend more money than you earn.

2. FAMILY SIZE AND COMPOSITION:

- A family of 7 will have a *much* larger food budget than a family of 2.
- The *composition* (ages) matters. A family with two teenagers (high energy needs) may spend more on food than a family with two infants.

3. FAMILY GOALS AND VALUES:

- What is important to the family?
- A family that *values* education will allocate a large amount to school fees.
- A family that *values* "Aso Ebi" and social events will budget for that.

4. LOCATION (RURAL VS. URBAN):

- An urban family (e.g., in Lagos) has higher costs for rent and food.
- A rural family may *grow* some of their own food, reducing their food budget.

5. MANAGEMENT SKILLS OF THE HOMEMAKER:

- A "skilled" manager will be able to make the money go further by budgeting, bulk-buying, and preventing waste.
- An "unskilled" manager will run out of money.

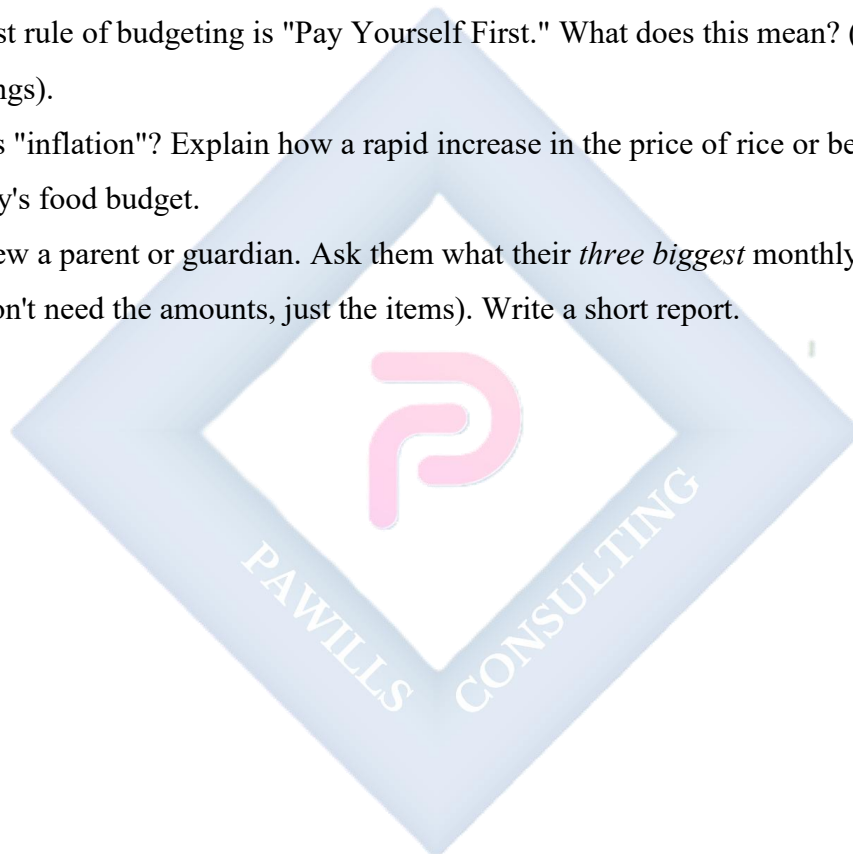
EVALUATION

1. What is a "budget"?
2. Explain the difference between "Income" and "Expenditure."
3. What is the difference between a "Fixed Expense" and a "Variable Expense"? Give one example of each.
4. Differentiate between a "Need" and a "Want."
5. List three factors that influence a family's budget.

ASSIGNMENT

1. Imagine you are a student given a monthly allowance of N10,000. Create a simple "Personal Budget" for one month.
 - **INCOME:** N10,000
 - **EXPENDITURE (PLAN):**

- Savings: N...?
 - Transport: N...?
 - Lunch/Snacks: N...?
 - Airtime/Data: N...?
 - Wants (e.g., cinema): N...?
 - **TOTAL EXPENSES:** (Must equal N10,000).
2. Write a 400-word essay on "Why a Food Budget is the most important part of a family's budget."
 3. The first rule of budgeting is "Pay Yourself First." What does this mean? (Hint: It relates to savings).
 4. What is "inflation"? Explain how a rapid increase in the price of rice or beans can "break" a family's food budget.
 5. Interview a parent or guardian. Ask them what their *three biggest* monthly expenses are (you don't need the amounts, just the items). Write a short report.



WEEK 9: FOOD PURCHASING

TOPIC: FOOD PURCHASING

INTRODUCTION: Food purchasing is the *action* part of the food budget. It is the practical skill of acquiring food for the household. The *way* you shop (your "purchasing habits") has a direct and immediate impact on the family's health and finances.



1: MEANING OF PURCHASING TERMS

- **BULK-PURCHASING:**

- **MEANING:** Buying food in large quantities (e.g., a whole bag of rice, a full basket of tomatoes, a whole cow shared with others).
- **WHEN:** Often done weekly, monthly, or seasonally (e.g., buying yams during the harvest).

- **WHERE:** Wholesale markets (e.g., Mile 12), supermarkets, or direct from a farm.
- **PIECE-MEAL PURCHASING:**
 - **MEANING:** Buying food in small quantities, as needed, often on a daily basis.
 - **WHEN:** (e.g., "I am cooking stew now, let me go and buy 3 tomatoes").
 - **WHERE:** Local shops, street vendors, neighbourhood markets.
- **IMPULSE BUYING:**
 - **MEANING:** Buying an item (usually a "want") that you did *not* plan to buy. It is an unplanned, spontaneous purchase.
 - **CAUSE:** Attractive displays (e.g., sweets at the checkout counter), "Buy One Get One Free" offers, or shopping when hungry.
 - **EFFECT:** It destroys a budget.

2: BULK-PURCHASING VS. PIECE-MEAL

FEATURE	BULK-PURCHASING	PIECE-MEAL PURCHASING
COST	Cheaper per unit (e.g., a bag of rice is cheaper per "derica" than buying one cup).	More expensive per unit.
TIME/ENERGY	Saves time and energy. One big trip instead of 10 small ones.	Wastes time and energy. Daily trips to the market.
STORAGE	Requires storage space (e.g., pantry, freezer) and good storage skills.	Requires no storage space. You buy and use immediately.
WASTAGE	High risk of spoilage if not stored correctly (e.g., the tomatoes rot).	Low risk of spoilage.
SUITABILITY	Best for non-perishable (rice, beans) and freezable (meat) foods.	Used for highly-perishable (fresh leafy greens) foods.
BUDGETING	Excellent for budgeting. You know your monthly food cost.	Bad for budgeting. It's easy to overspend.

3: FACTORS TO CONSIDER WHEN PURCHASING FOOD

1. **THE BUDGET:** This is the *first* consideration. How much money do you have?
2. **THE SHOPPING LIST:**
 - **MEANING:** A list, written *before* going to the market, of all the items you need.
 - **IMPORTANCE:** This is the **number one tool** against "impulse buying." It keeps you focused and ensures you don't forget essentials.
3. **THE MENU / MEAL PLAN:** Your list should be based on your *menu* for the week. (e.g., "I am making Jollof rice, so I need rice, tomatoes, oil...").
4. **NUTRITIONAL VALUE:** Choosing items that meet the family's health needs (from Week 4).
5. **STORAGE FACILITIES:** (From Week 4) - Do not buy a crate of fish if your freezer is broken.
6. **FRESHNESS & QUALITY:** (From Term 1) - Checking the fish gills, the can dents, etc.
7. **THE SEASON:** Buying "in season" (e.g., mangoes, yams at harvest) is *always* cheaper and the quality is higher.

EVALUATION

1. Explain the difference between "Bulk-Purchasing" and "Piece-Meal Purchasing."
2. What is "Impulse Buying," and what is the single best tool to prevent it?
3. List three *advantages* of bulk-purchasing.
4. List two *disadvantages* of bulk-purchasing.
5. What is the relationship between a "Menu Plan" and a "Shopping List"?

ASSIGNMENT

1. Create a "Shopping List" for your family for one week. Base it on a simple menu plan (e.g., "Jollof Rice on Sunday," "Beans on Tuesday").
2. Write a 400-word essay arguing *for* or *against* this statement: "Bulk-purchasing is always better than piece-meal purchasing."
3. Where in a supermarket are "impulse buys" (like chocolate, magazines) usually placed, and *why* are they placed there?

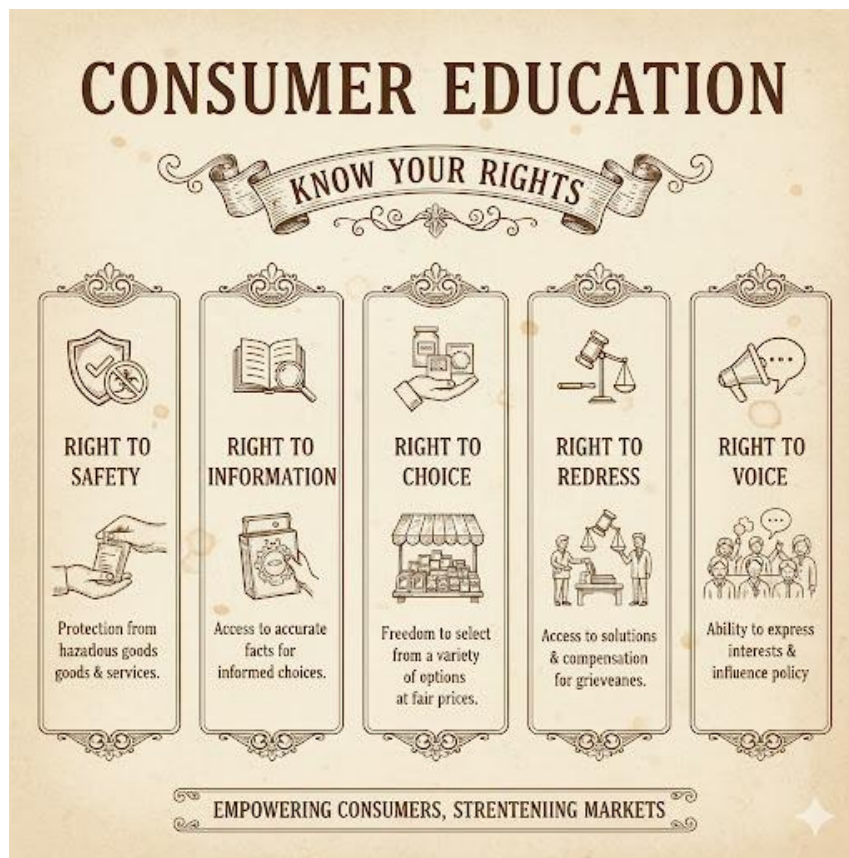
4. What does "buying in season" mean? List two food items that are "in season" in Nigeria *right now* (or in the current local season).
5. Research the "pros and cons" of "shopping at an open-air market" vs. "shopping at a modern supermarket." Create a two-column comparison table.



WEEK 10: CONSUMER EDUCATION I & II

TOPIC: CONSUMER EDUCATION AND PROTECTION

INTRODUCTION: A "consumer" is any person who buys goods or uses services. "Consumer Education" is the process of teaching consumers their *rights* and *responsibilities* in the marketplace, empowering them to make wise, informed, and safe choices.



1: MEANING AND PRINCIPLES OF CONSUMER EDUCATION

- **MEANING:** Consumer education is the preparation of an individual to be a skillful and informed consumer, capable of protecting themselves from exploitation and making choices that improve their quality of life.
- **PRINCIPLES (THE "RIGHTS" OF A CONSUMER):**

1. **THE RIGHT TO SAFETY:** To be protected from goods and services that are hazardous to health or life (e.g., expired drugs, unpasteurized milk).
2. **THE RIGHT TO BE INFORMED:** To be given the *facts* (e.g., labels, ingredients, price) needed to make an informed choice.
3. **THE RIGHT TO CHOOSE:** To have access to a *variety* of goods and services at competitive prices.
4. **THE RIGHT TO BE HEARD:** To have your complaints and interests listened to by businesses and government.
5. **THE RIGHT TO REDRESS (COMPENSATION):** To receive fair compensation (e.g., a refund, replacement) for a faulty good or bad service.

2: COMMON UNFAIR CONSUMER PRACTICES

Consumers must be educated to protect themselves from these common bad practices:

1. ADULTERATION:

- **MEANING:** Adding a *cheaper*, inferior, or harmful substance to a food to increase its quantity and *profit*.
- **EXAMPLES:**
 - Adding water to milk.
 - Adding *stones* or *sand* to rice or beans to increase the weight.
 - Using *Sudan Red* (a toxic dye) to make palm oil look redder.

2. FALSIFICATION (MISBRANDING / MISLABELING):

- **MEANING:** Lying on the label or packaging.
- **EXAMPLES:**
 - Putting a *fake expiration date* on a product.
 - Writing "100% Fruit Juice" when it is only 10% juice and 90% sugar water.
 - Selling "local rice" in a bag labelled "imported rice" for a higher price.
 - **Falsifying weight:** (e.g., the "mudu" or "derica" cup has a thick bottom, so it holds less).

3. HOARDING:

- **MEANING:** When a seller hides a large stock of an essential product (e.g., fuel, rice) to create an *artificial scarcity*, which then drives up the price.
- 4. **SELLING EXPIRED OR UNSAFE GOODS:** Operating an unhygienic restaurant or selling food past its "Use By" date.

3: GOVERNMENT CONSUMER AGENTS (PROTECTION)

The government has created agencies to enforce consumer rights and punish bad practices.

1. NAFDAC (National Agency for Food and Drug Administration and Control):

- **JURISDICTION:** Regulates *processed* food, drugs, cosmetics, and water.
- **DUTIES:**
 - "Safeguarding public health."
 - **Inspects** factories and ports.
 - **Tests** products in labs.
 - **Issues "NAFDAC Numbers"** - A NAFDAC number on a product (e.g., "sachet water," bread) signifies it has been tested and is registered as safe.
 - **Fights** against fake drugs and adulterated food.

2. SON (Standards Organisation of Nigeria):

- **JURISDICTION:** Regulates *all* manufactured products (not just food/drugs).
- **DUTIES:**
 - **Sets the "standard"** for quality (e.g., "What is the standard for a safe car tyre?").
 - **Monitors** product quality.
 - **Issues "MANCAP"** (Mandatory Conformity Assessment Programme) certification.

3. FCCPC (Federal Competition and Consumer Protection Commission):

- **JURISDICTION:** This is the *main* consumer "watchdog."
- **DUTIES:**
 - **Fights** for consumer *rights* (from 1).
 - **Handles** consumer *complaints* (e.g., "My new DSTV decoder didn't work, and the company refused a refund").

- **Ensures** "fair competition" (e.g., stops one company from having an unfair monopoly).
- **Can sue** companies on behalf of consumers.

EVALUATION

1. What is a "consumer"?
2. List the five main "Rights of a Consumer."
3. What is "adulteration"? Give one specific example.
4. What is "falsification" (misbranding)? Give one specific example.
5. What is the full meaning of **NAFDAC**, and what is its primary duty?

ASSIGNMENT

1. Find three *different* processed food items in your home that have a "NAFDAC Number." Write down the product name and the number.
2. What is the "Right to Redress"? Write a 400-word story about a consumer who buys a faulty blender, what steps they should take, and what the final (fair) outcome should be.
3. Research the "Sudan Red" palm oil scandal. Write a one-paragraph summary explaining what it was and which *consumer right* it violated.
4. What is the difference between NAFDAC and SON? Create a two-column table to compare their main duties.
5. Find the official website or social media handle for the FCCPC. What is one *recent case* or *warning* they have issued to the public?

WEEK 11: REVISION

TOPIC: GENERAL REVIEW OF ALL TOPICS

INTRODUCTION: This week is dedicated to a comprehensive review of all topics covered in SS2 Third Term. We will consolidate our knowledge on food processing, storage, and consumer economics to prepare for the final examinations.

REVISION SCOPE:

- **FOOD PROCESSING & STORAGE:**
 - **Convenience Foods:** Forms, pros (time), cons (salt, cost, nutrition).
 - **Spoilage:** Agents (microbes, enzymes), principles of preservation (remove water/air, kill microbes, slow growth).
 - **Preservation Methods:** Canning, Freezing, Drying, Salting, Smoking, Pickling, Fermentation.
 - **Storage:** Perishables ("raw below cooked"), Non-perishables (airtight, FIFO), Equipment (fridge, freezer, barn).
- **FOOD ECONOMICS & MANAGEMENT:**
 - **Budgeting:** Income, Expenditure (fixed/variable), Needs vs. Wants, factors (family size).
 - **Purchasing:** Bulk vs. Piece-meal, Impulse buying, Shopping lists.
 - **Consumer Education:** Rights (safety, info), Bad Practices (adulteration, falsification), Agencies (NAFDAC, SON, FCCPC).

REVISION STRATEGY:

1. **Review all notes.**
2. **Answer all "Evaluation" questions** from each week.
3. **PRACTICE ESSAY QUESTIONS:**
 - "Discuss the *advantages* and *disadvantages* of the widespread use of convenience foods in modern society."

- "You are a consumer rights advocate. Educate the public on five common *unfair practices* and the three main *government agencies* that can protect them."
- "A family's income has been cut by 30%. Advise them on how to manage their *food budget* using smart *purchasing habits* and *preservation methods*."
- "Explain the scientific principles behind these five preservation methods: (a) Freezing, (b) Salting, (c) Canning, (d) Pickling, (e) Sun-drying."
- "What is the 'raw below cooked' rule, and why is it the most important principle of food storage in a refrigerator? Discuss."

